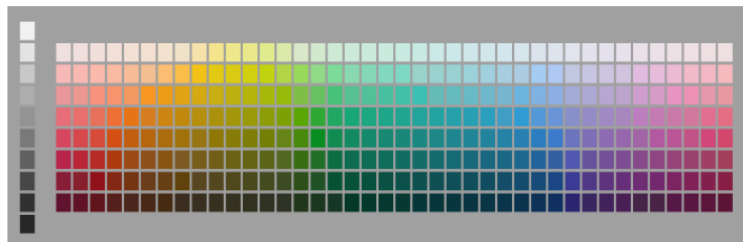
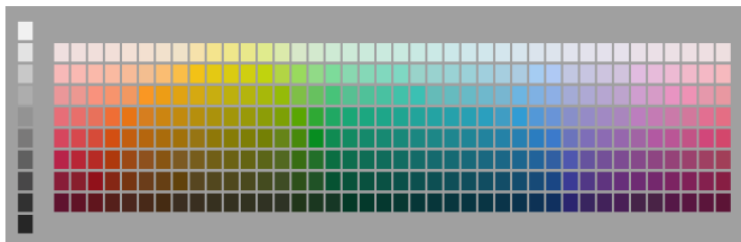


Boas, Shannon, and the origin of semantic categories

Terry Regier

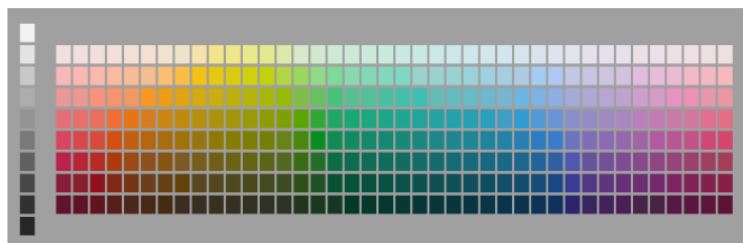
SIGTYP 2026 at EACL 2026
Rabat, Morocco, March 29 2026





Wobé (Ivory Coast)



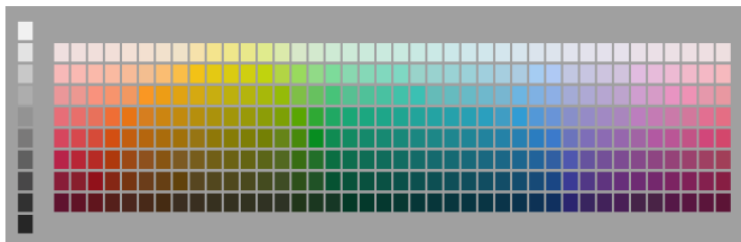


Wobé (Ivory Coast)



Buglere (Panama)





Semantic typology

Why do we find the category systems we do in the world's languages?

Wobé (Ivory Coast)

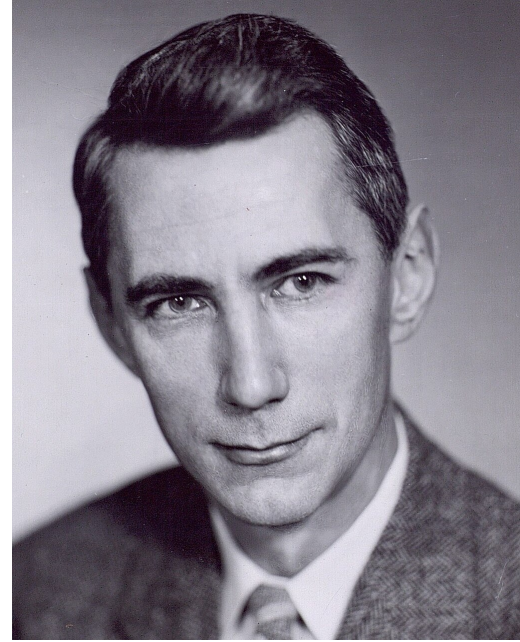


Buglere (Panama)

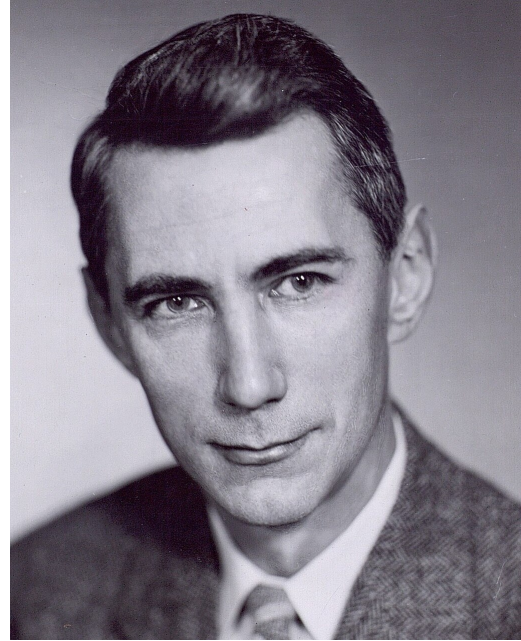




Franz Boas (1858-1942)
Father of American anthropology



Claude Shannon (1916-2001)
Father of information theory



Claude Shannon (1916-2001)
Father of information theory

INFORMATION

SOURCE

TRANSMITTER

RECEIVER

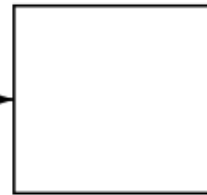
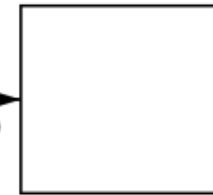
DESTINATION



SIGNAL

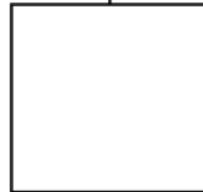


RECEIVED
SIGNAL



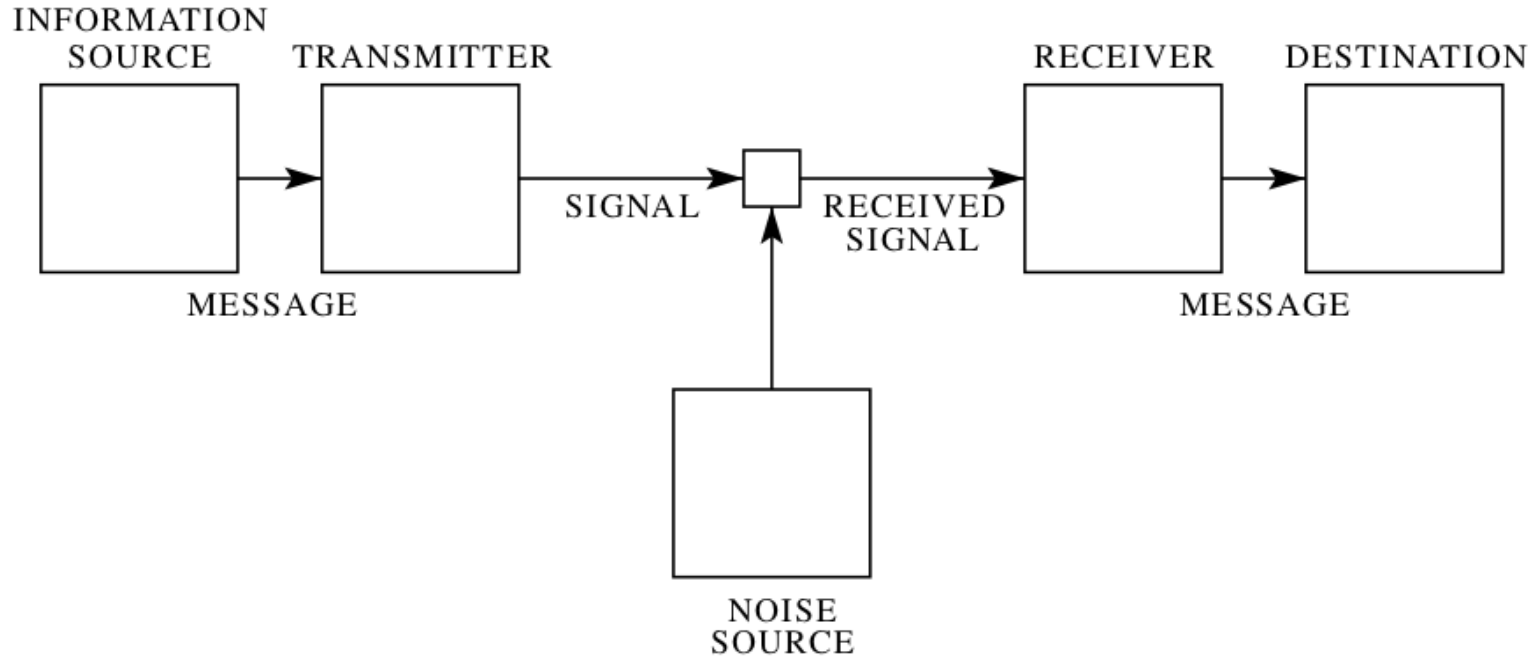
MESSAGE

MESSAGE

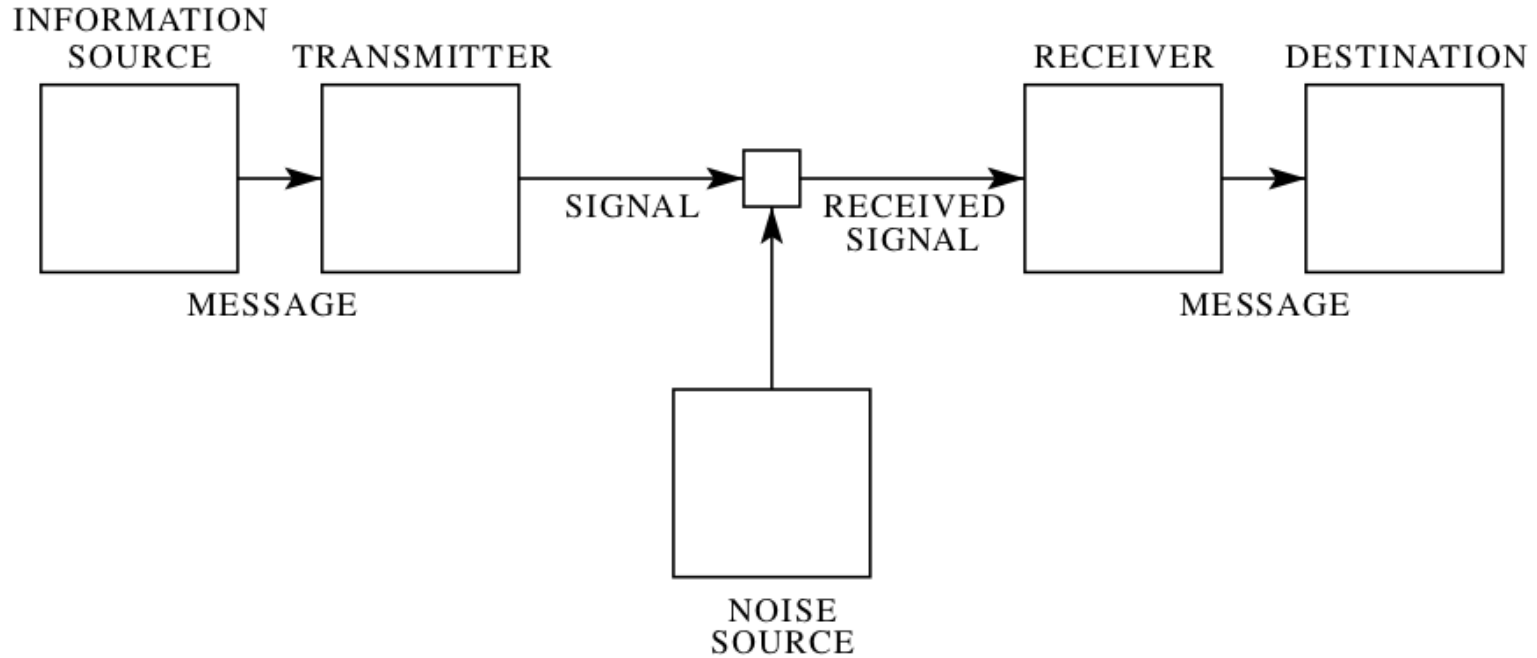


NOISE
SOURCE





“Frequently the messages have **meaning**; [t]hese **semantic** aspects of communication are **irrelevant** to the engineering problem.” (Shannon, 1948)



Information theory is “**not necessarily relevant** to such fields as psychology, economics, and other **social sciences.**” (Shannon, 1956)

Efficient compression in color naming and its evolution

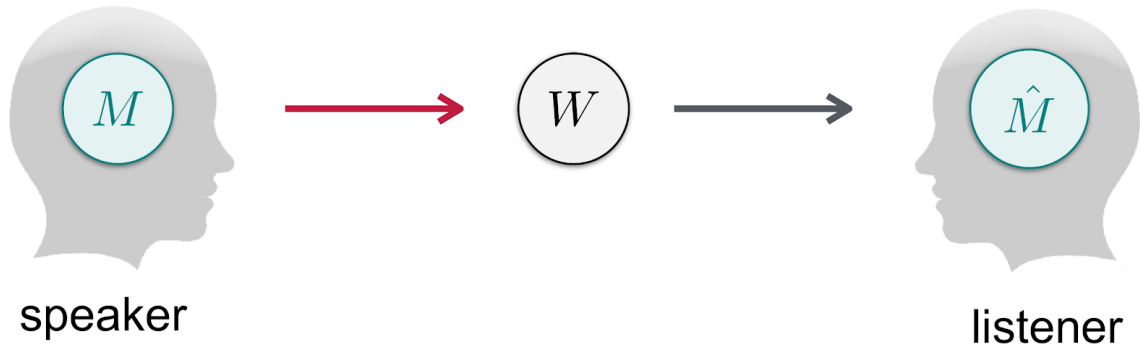
Noga Zaslavsky^{a,b,1}, Charles Kemp^{c,2}, Terry Regier^{b,d}, and Naftali Tishby^{a,e}

Proceedings of the National Academy of Sciences, 2018

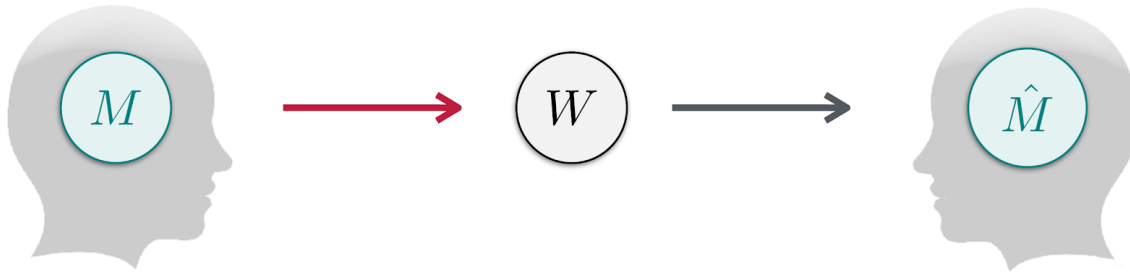


Noga Zaslavsky

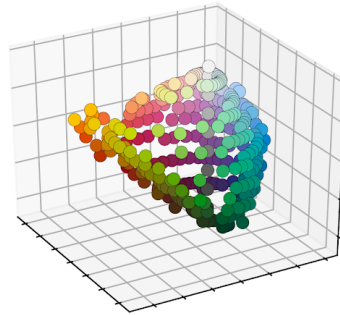
Communication model



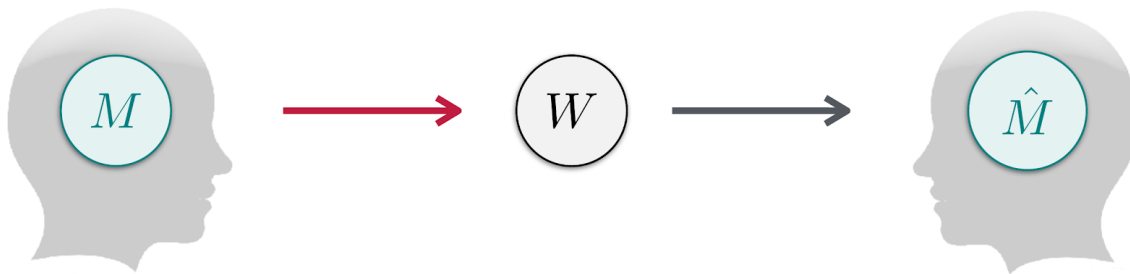
Communication model



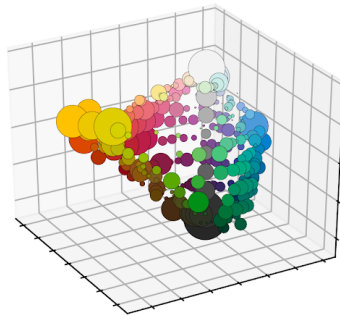
- Perceptually grounded meanings
[Regier et al., 2007, 2015]



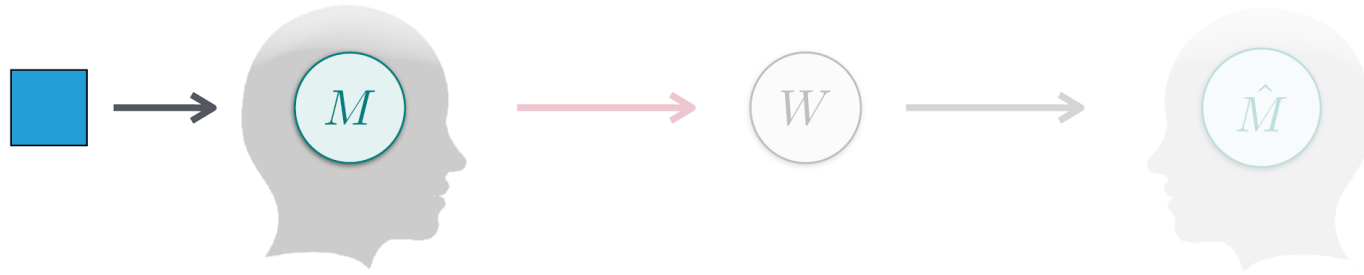
Communication model



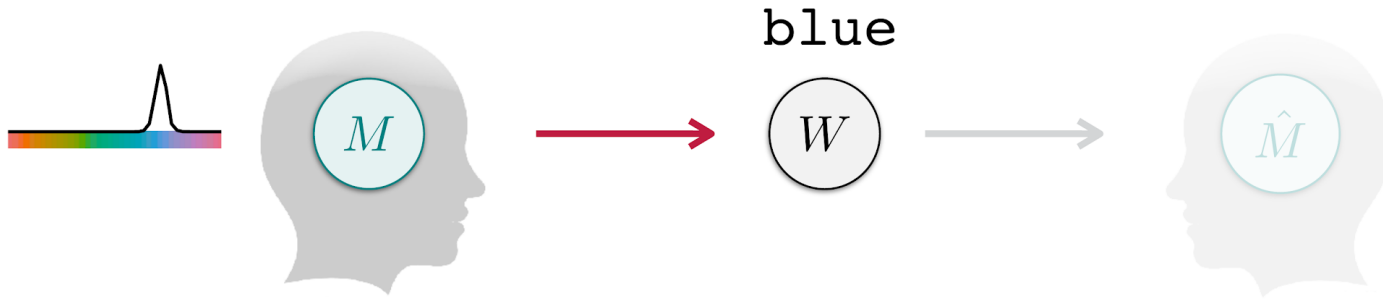
- Perceptually grounded meanings
[Regier et al., 2007, 2015]
- Communicative need
[Zaslavsky et al., 2018]



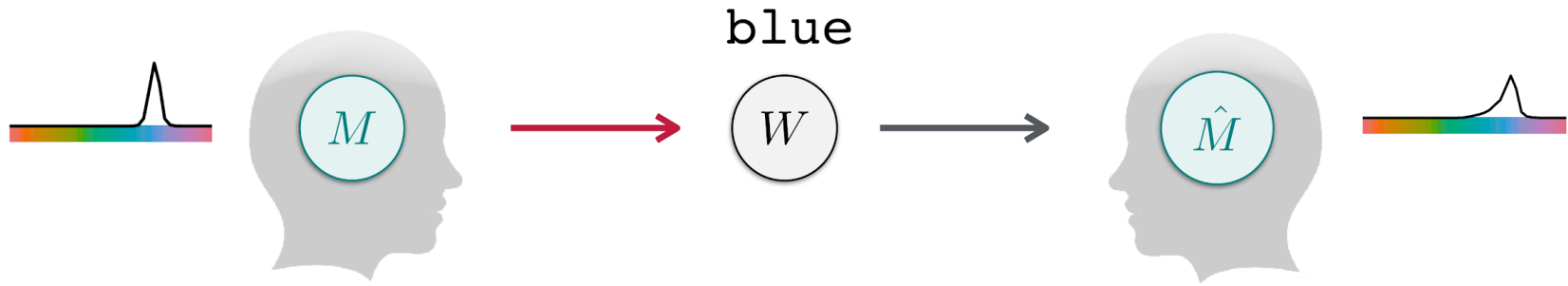
Communication model



Communication model

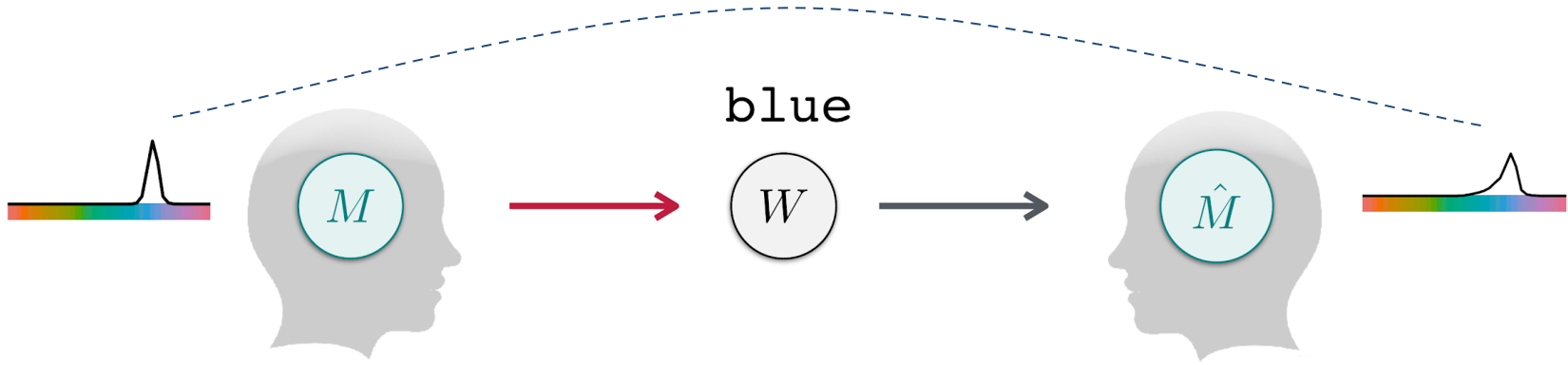


Communication model

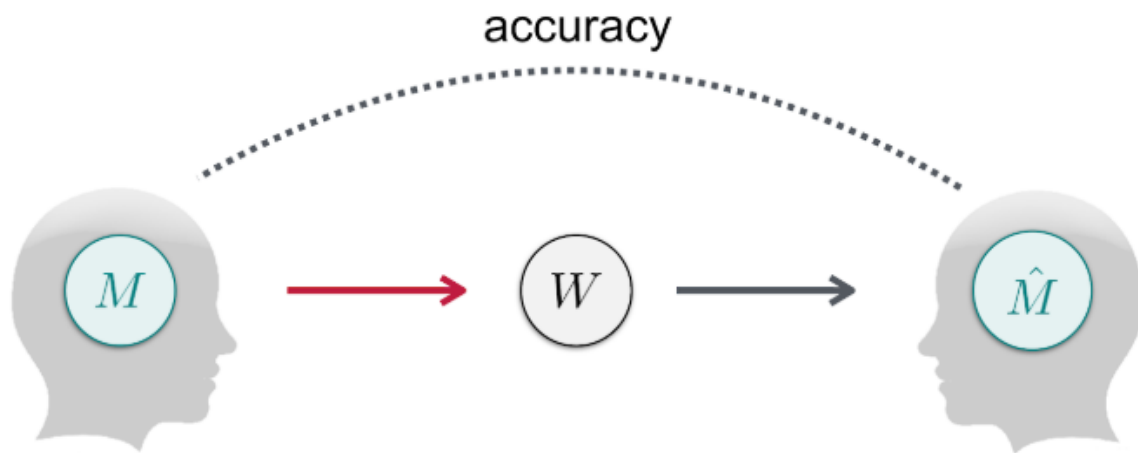


Communication model

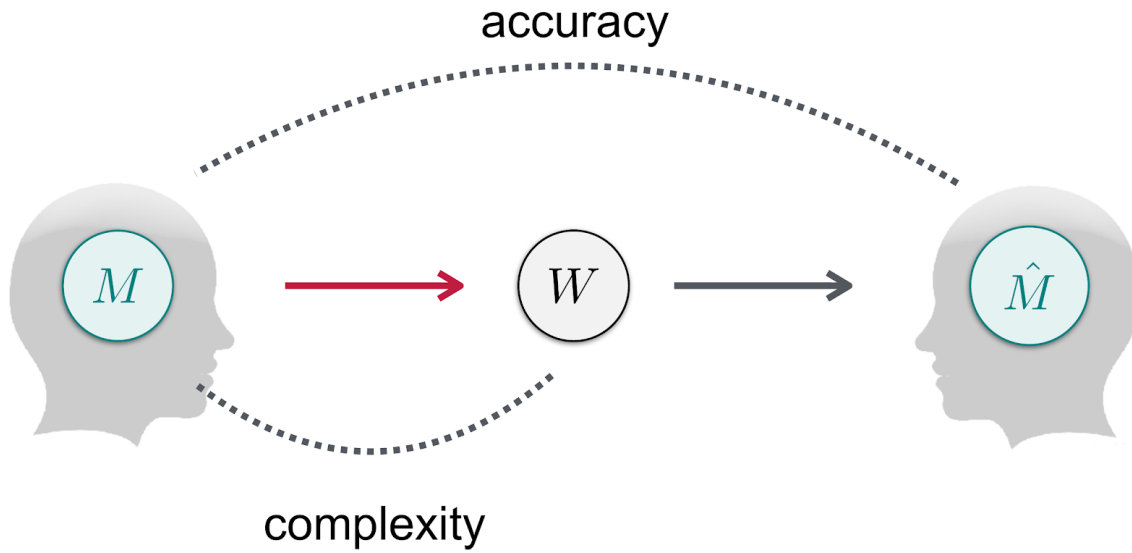
information loss, or **inaccuracy**



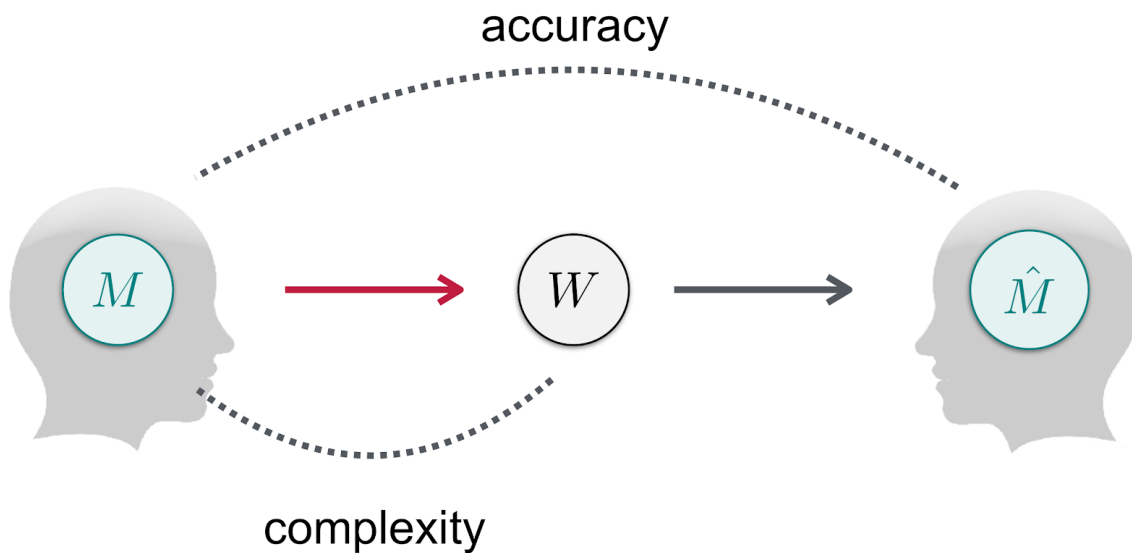
Efficient communication



Efficient communication



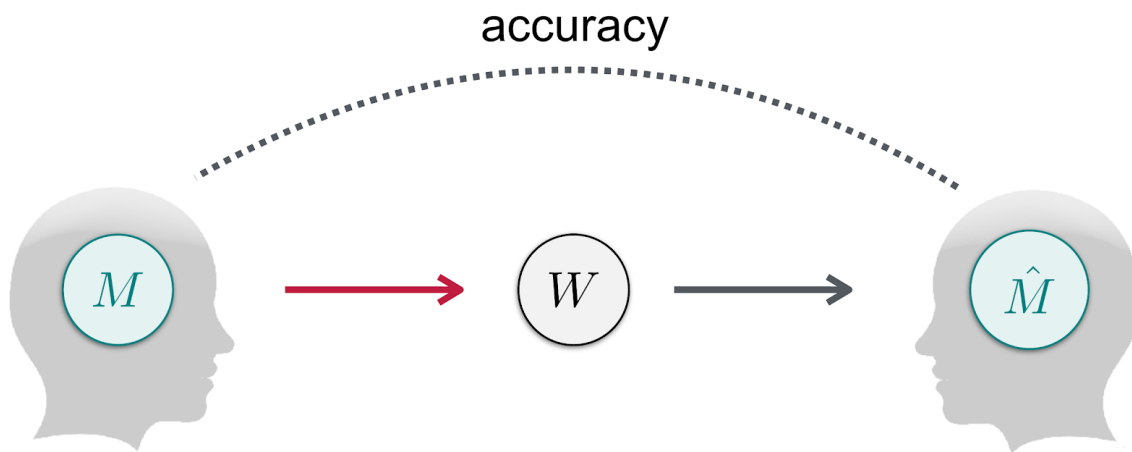
The Information Bottleneck (IB) principle



Tishby et al., 1999

Zaslavsky et al., *PNAS*, 2018

The Information Bottleneck (IB) principle

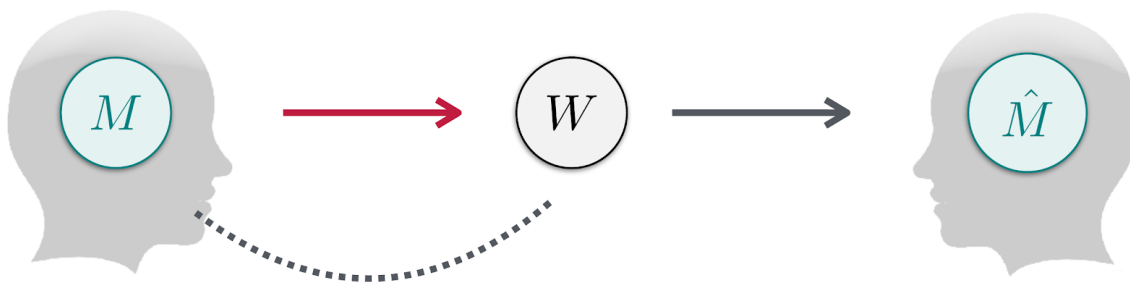


High accuracy means low distortion, $\mathbb{E}[D[M||\hat{M}]]$

Tishby et al., 1999

Zaslavsky et al., *PNAS*, 2018

The Information Bottleneck (IB) principle

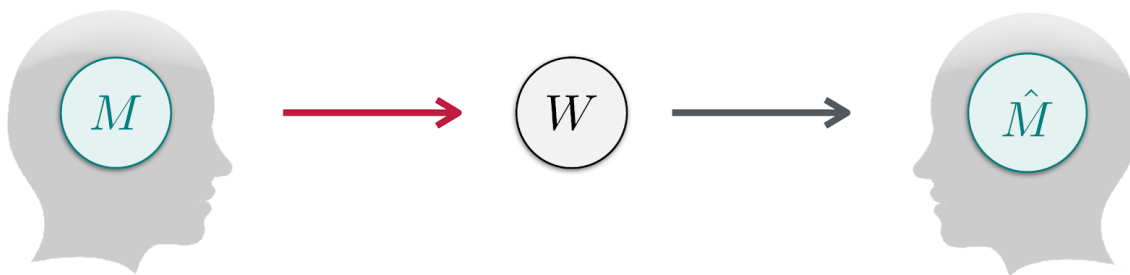


$$\text{complexity} = I(M; W)$$

Tishby et al., 1999

Zaslavsky et al., *PNAS*, 2018

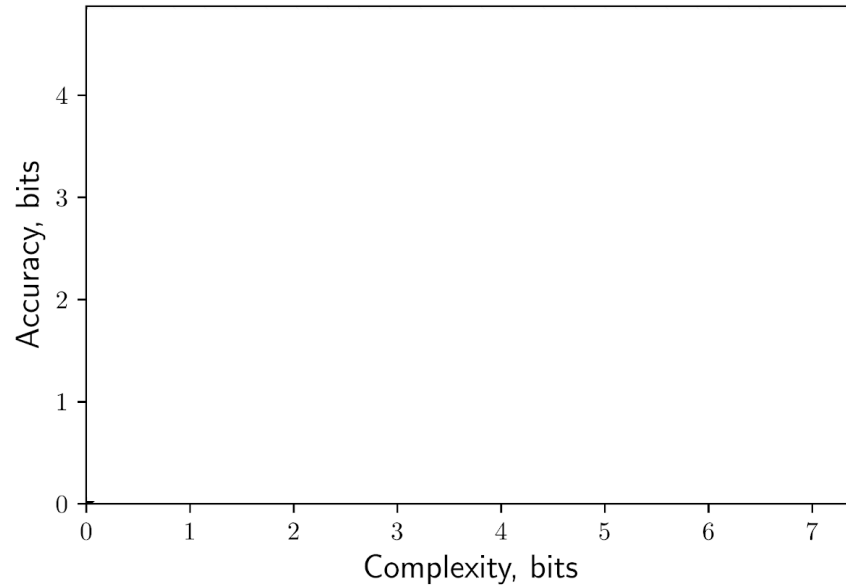
The Information Bottleneck (IB) principle



Optimal **systems** minimize: **complexity** $-\beta$ **accuracy**

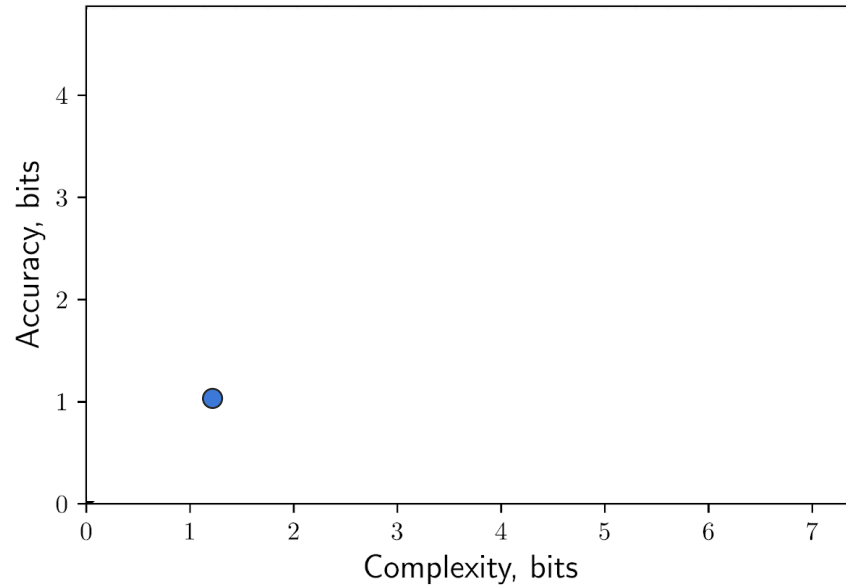
↑
tradeoff

The information plane

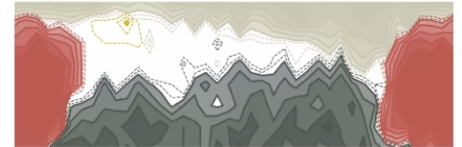


Zaslavsky et al., *PNAS*, 2018

The information plane

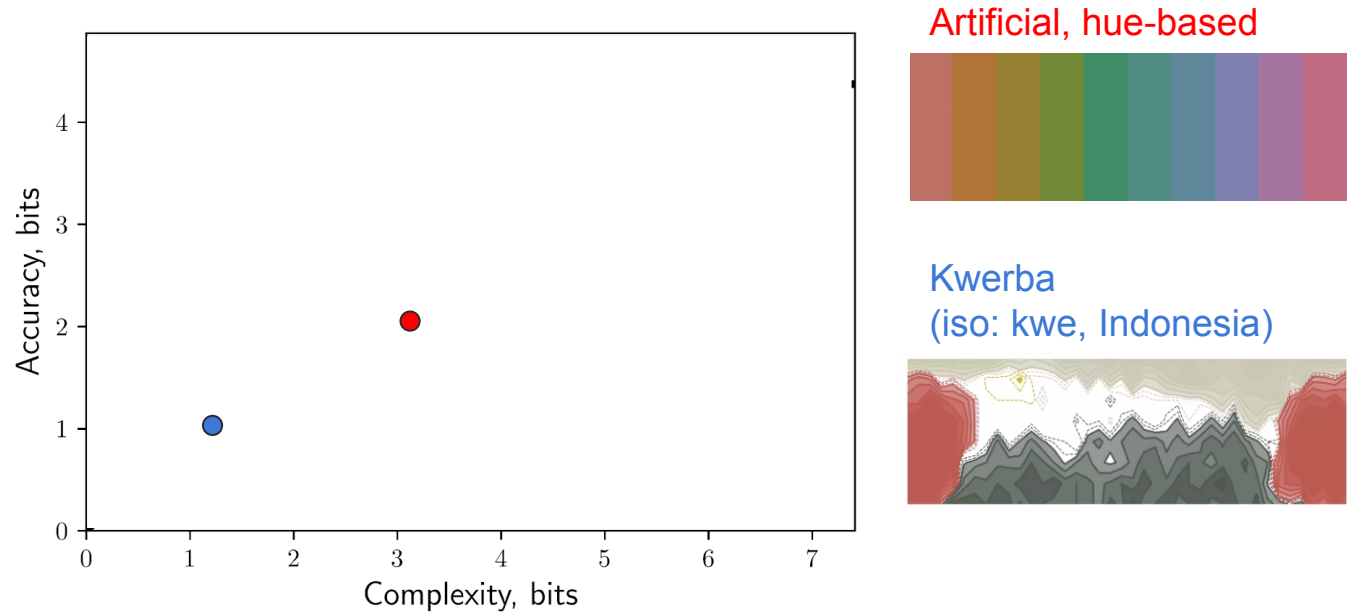


Kwerba
(iso: kwe, Indonesia)



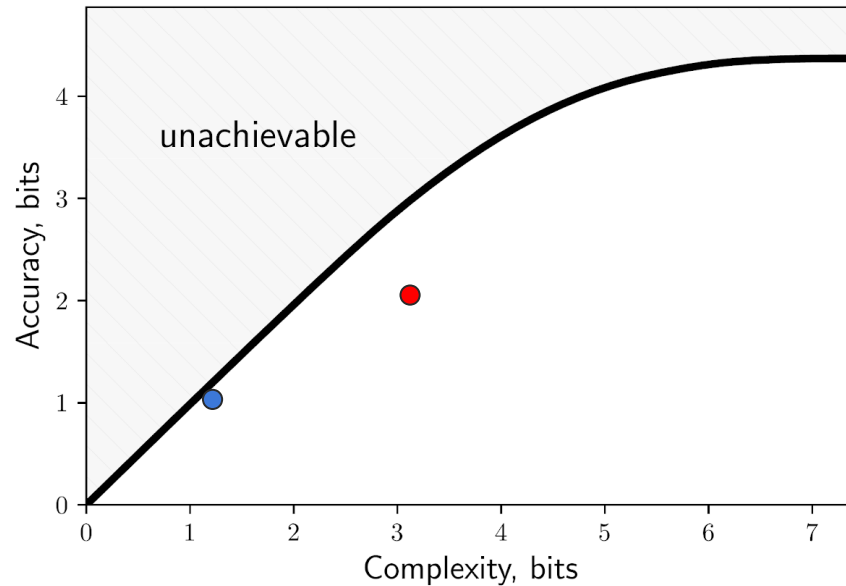
Zaslavsky et al., *PNAS*, 2018

The information plane



Zaslavsky et al., *PNAS*, 2018

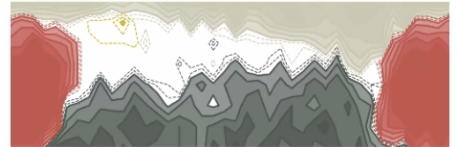
The information plane



Artificial, hue-based

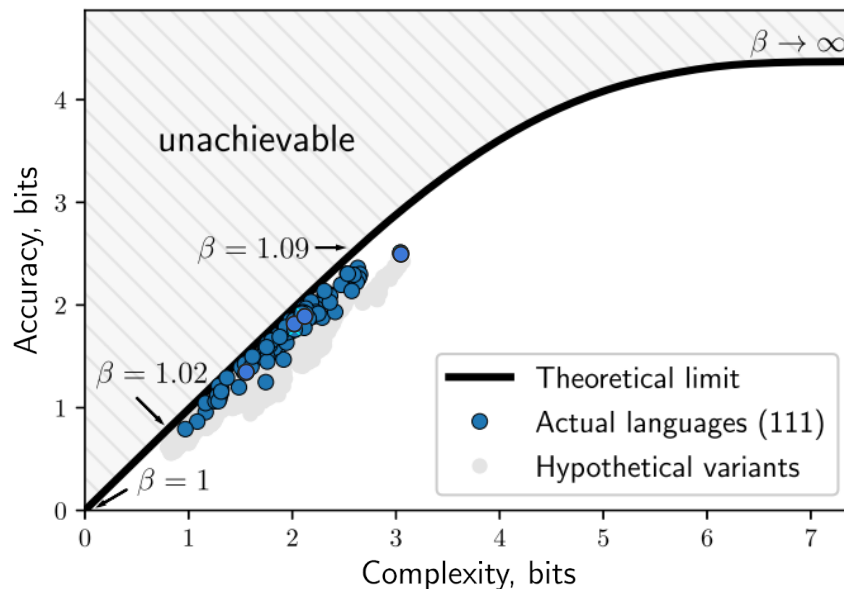


Kwerba
(iso: kwe, Indonesia)



Zaslavsky et al., *PNAS*, 2018

Languages are near-optimally efficient



110 languages
of the World
Color Survey,
+ English

Explaining known descriptive generalizations

$\begin{bmatrix} \text{white} \\ \text{black} \end{bmatrix} < [\text{red}] < \begin{bmatrix} \text{green} \\ \text{yellow} \end{bmatrix} < [\text{blue}] < [\text{brown}] < \begin{bmatrix} \text{purple} \\ \text{pink} \\ \text{orange} \\ \text{grey} \end{bmatrix}$

Berlin & Kay 1969

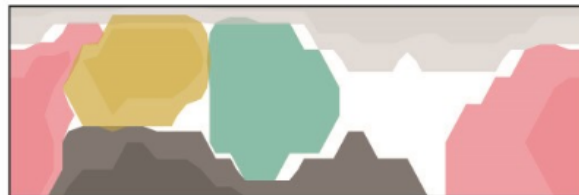
Explaining known descriptive generalizations

$$\begin{bmatrix} \text{white} \\ \text{black} \end{bmatrix} < [\text{red}] < \begin{bmatrix} \text{green} \\ \text{yellow} \end{bmatrix} < [\text{blue}] < [\text{brown}] < \begin{bmatrix} \text{purple} \\ \text{pink} \\ \text{orange} \\ \text{grey} \end{bmatrix}$$

Berlin & Kay 1969

while addressing a known critique

Optimal:



Levinson 2001

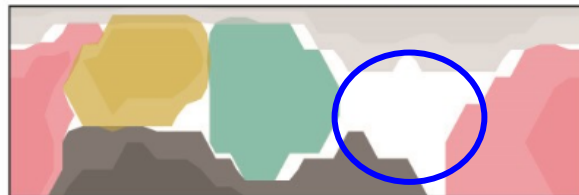
Explaining known descriptive generalizations

$$\begin{bmatrix} \text{white} \\ \text{black} \end{bmatrix} < [\text{red}] < \begin{bmatrix} \text{green} \\ \text{yellow} \end{bmatrix} < [\text{blue}] < [\text{brown}] < \begin{bmatrix} \text{purple} \\ \text{pink} \\ \text{orange} \\ \text{grey} \end{bmatrix}$$

Berlin & Kay 1969

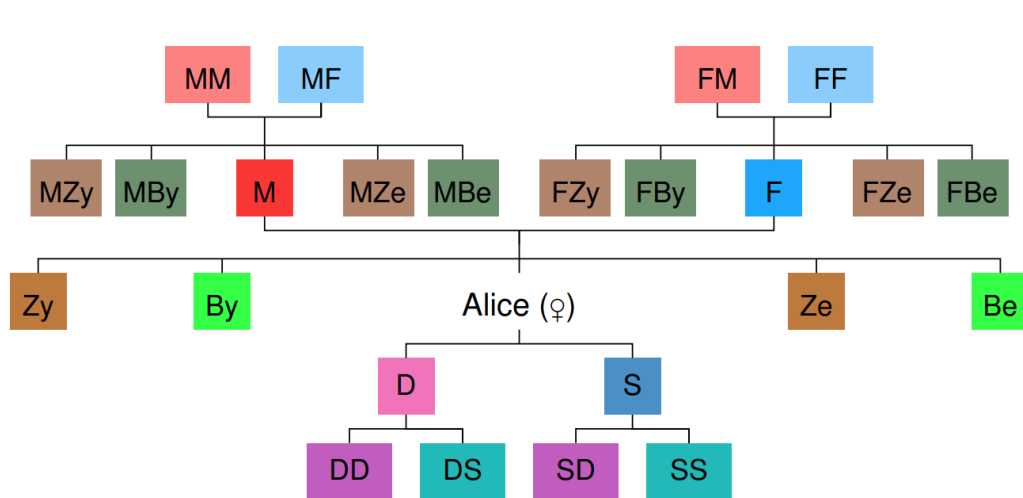
while addressing a known critique

Optimal:



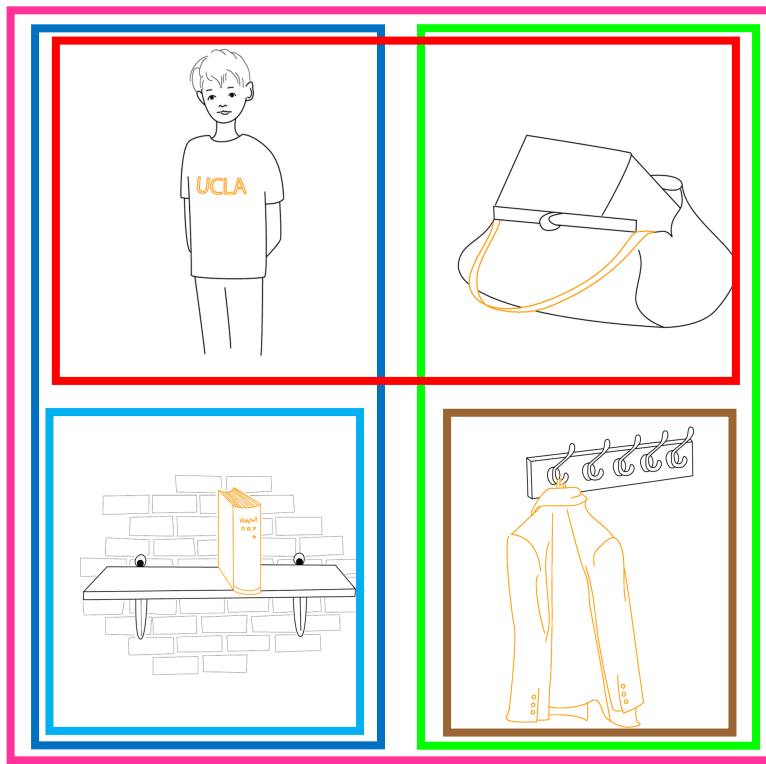
Levinson 2001

Kinship



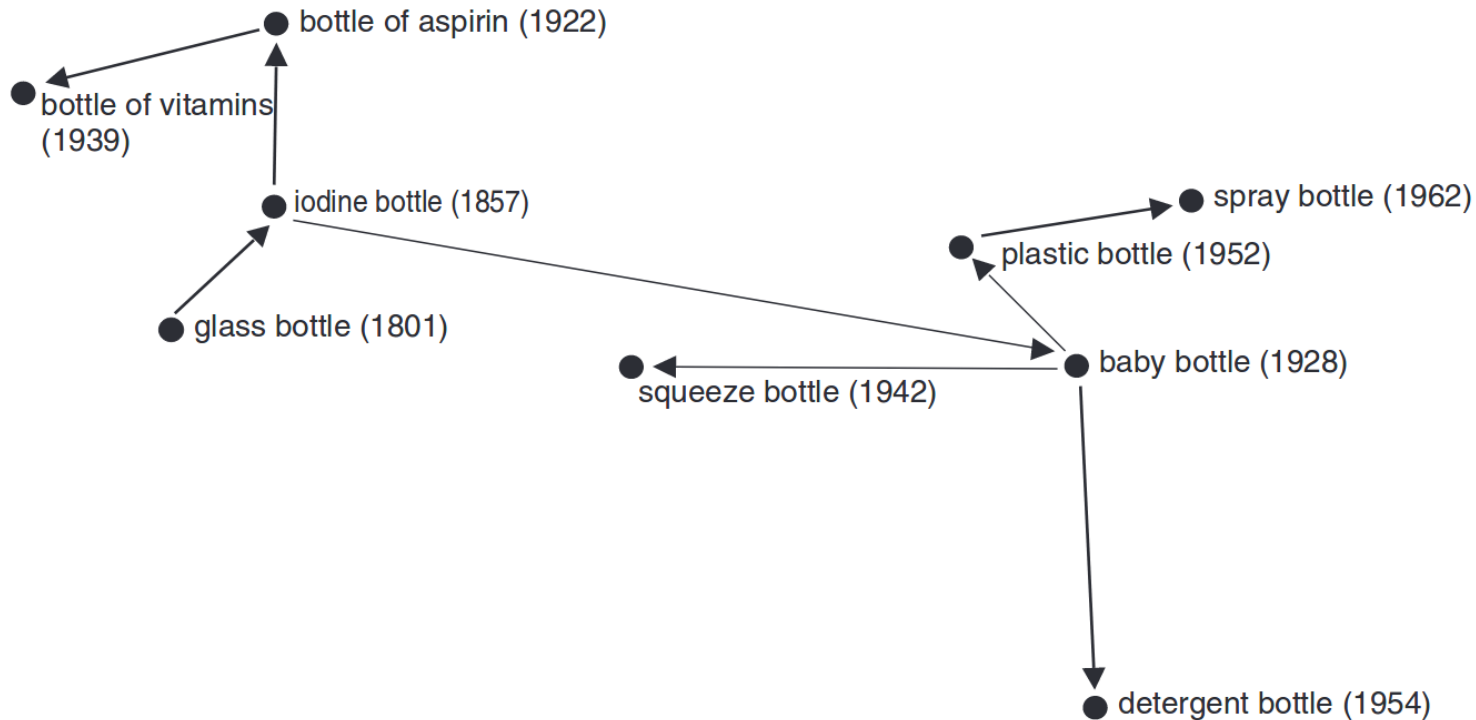
■ mother (x, y)	$\leftrightarrow \text{PARENT}(x, y) \wedge \text{FEMALE}(x)$
■ father (x, y)	$\leftrightarrow \text{PARENT}(x, y) \wedge \text{MALE}(x)$
■ daughter (x, y)	$\leftrightarrow \text{CHILD}(x, y) \wedge \text{FEMALE}(x)$
■ son (x, y)	$\leftrightarrow \text{CHILD}(x, y) \wedge \text{MALE}(x)$
■ sister (x, y)	$\leftrightarrow \exists z \text{ daughter}(x, z) \wedge \text{PARENT}(z, y)$
■ brother (x, y)	$\leftrightarrow \exists z \text{ son}(x, z) \wedge \text{PARENT}(z, y)$
sibling (x, y)	$\leftrightarrow \exists z \text{ CHILD}(x, z) \wedge \text{PARENT}(z, y)$
■ aunt (x, y)	$\leftrightarrow \exists z \text{ sister}(x, z) \wedge \text{PARENT}(z, y)$
■ uncle (x, y)	$\leftrightarrow \exists z \text{ brother}(x, z) \wedge \text{PARENT}(z, y)$
■ niece (x, y)	$\leftrightarrow \exists z \text{ daughter}(x, z) \wedge \text{sibling}(z, y)$
■ nephew (x, y)	$\leftrightarrow \exists z \text{ son}(x, z) \wedge \text{sibling}(z, y)$
■ grandmother (x, y)	$\leftrightarrow \exists z \text{ mother}(x, z) \wedge \text{PARENT}(z, y)$
■ grandfather (x, y)	$\leftrightarrow \exists z \text{ father}(x, z) \wedge \text{PARENT}(z, y)$
■ granddaughter (x, y)	$\leftrightarrow \exists z \text{ daughter}(x, z) \wedge \text{CHILD}(z, y)$
■ grandson (x, y)	$\leftrightarrow \exists z \text{ son}(x, z) \wedge \text{CHILD}(z, y)$

Spatial terms



Khetarpal et al. 2013, *Cognitive Science conference*

Household containers



Xu et al. 2016, *Cognitive Science*

Numeral systems

English 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15



Yidiny 1 2 3 4 5 6 ... 15



Munduruku 1 2 3 4 5 6 ... 11 ...



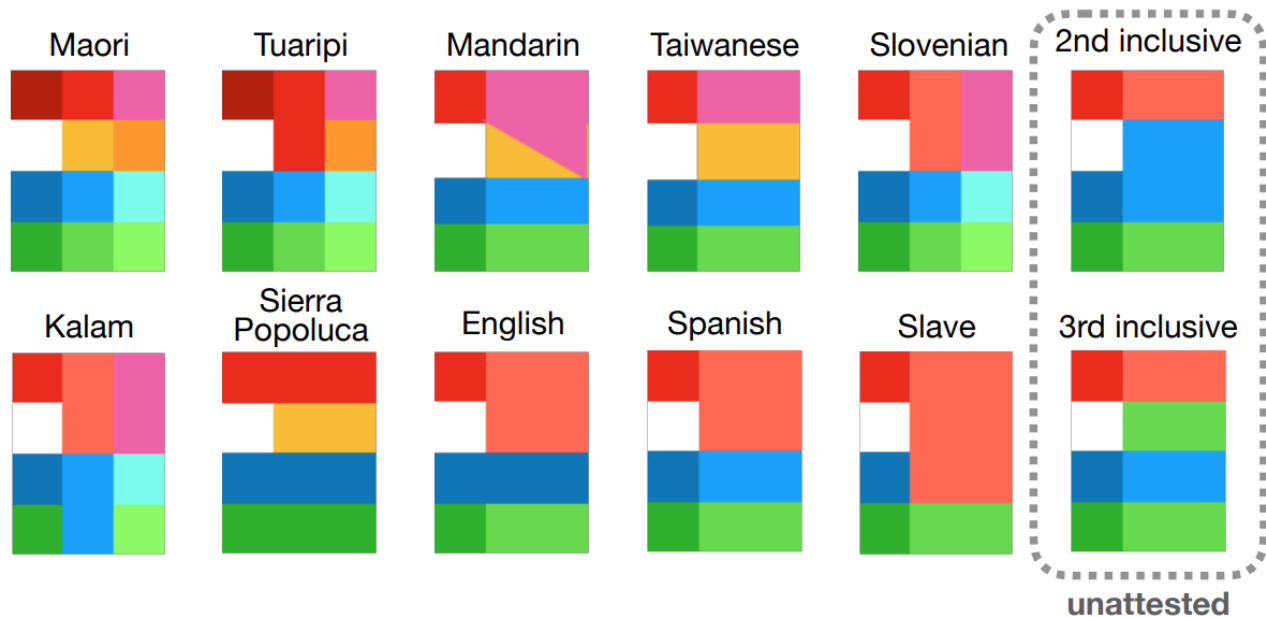
Xu et al. 2020, *Open Mind*;
but see also Denić & Szymanik 2024, *Cognitive Science*

Personal pronouns

A.

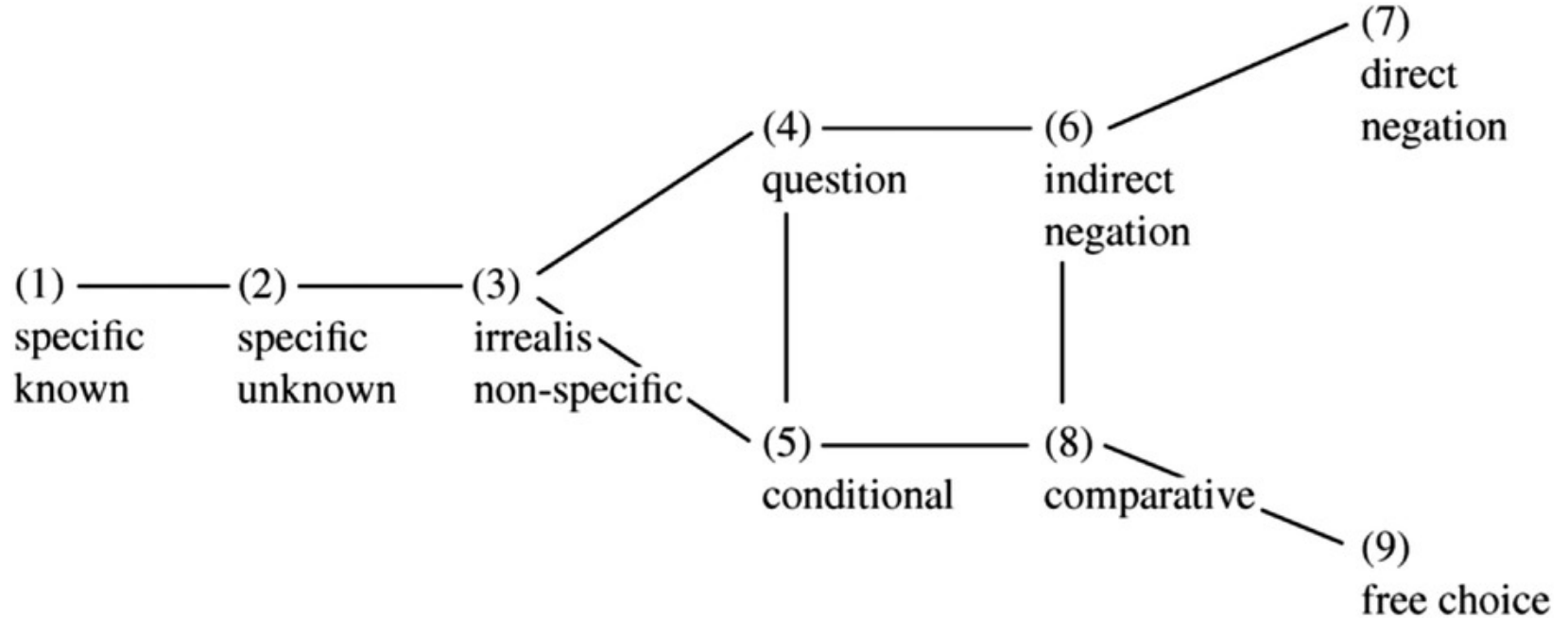
person	s	s+o	s+o+o ...
		s+a	s+a+o ...
	a	a+o	a+o+o ...
	o	o+o	o+o+o ...
number			

B.



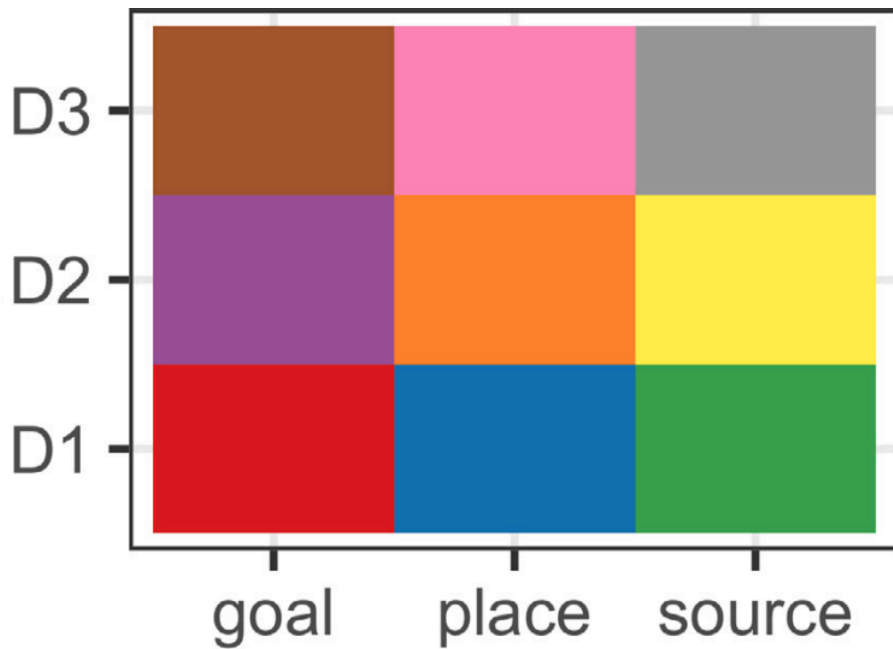
Zaslavsky et al. 2021, *Cognitive Science conference*

Indefinite pronouns



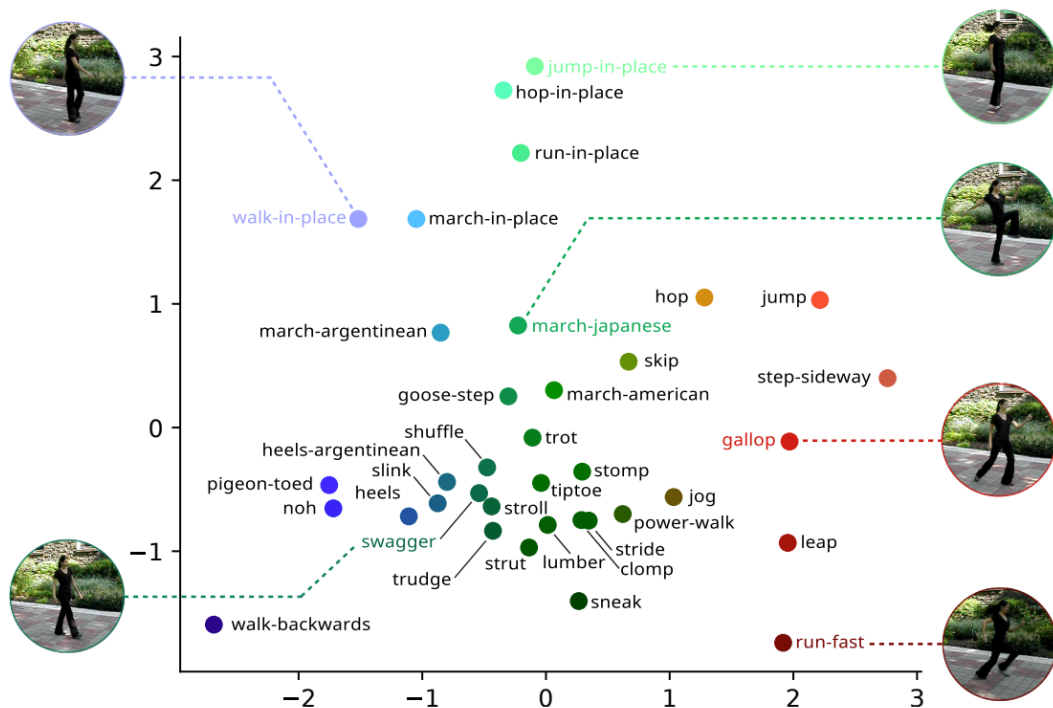
Denić et al. 2022, *Cognitive Science*

Spatial demonstratives



Chen et al. 2023, *Cognition*

Locomotion verbs



Langlois et al. 2025, *Cognitive Science conference*

Interim conclusions

- Semantic systems across languages and across domains may reflect **functional adaptation** in support of **efficient communication**.

Interim conclusions

- Semantic systems across languages and across domains may reflect **functional adaptation** in support of **efficient communication**.
- In line with Shannon's original model, **despite** his apparent skepticism.



Franz Boas (1858-1942)
Father of American anthropology





Among “the words for snow in Eskimo [Inuit] ... we find one word, *aput*, expressing snow on the ground; another one, *qana*, falling snow; ...”

Franz Boas, 1911



“It seems fairly evident that the selection of such simple terms must to a certain extent depend upon **the chief interests of a people**”

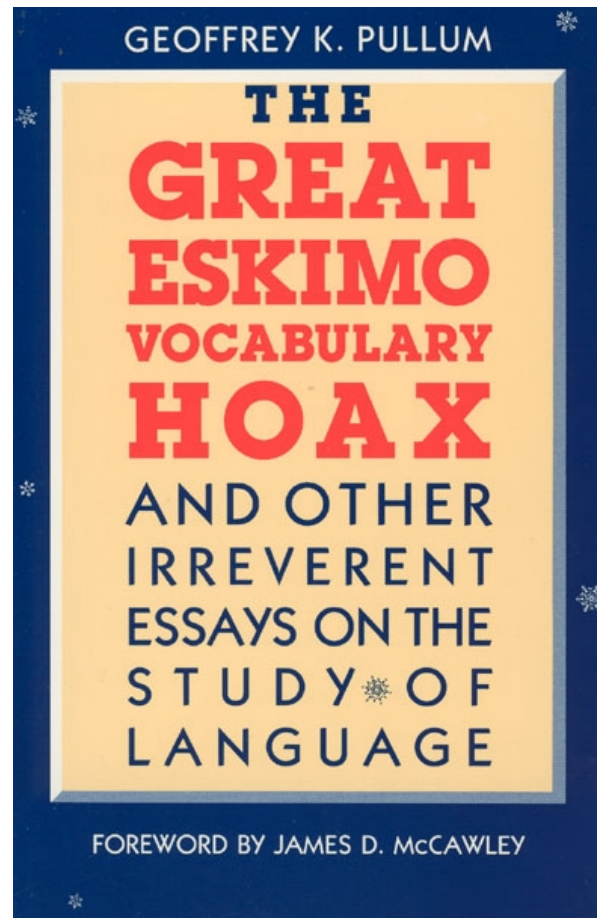
Franz Boas, 1911



Benjamin Lee Whorf, 1956

“unredeemed piffle”

Geoffrey Pullum, 1991



Geoffrey Pullum, 1991

If the Eskimos have 500 words for snow, then surely we in our internet-stuff age should have 1000 words for "meh"!



T-REX THAT LINGUISTIC PREMISE IS ACTUALLY FULL OF LIES

gasp



Dinosaur Comics, 2011

This bothered me

INFORMATION

SOURCE

TRANSMITTER

RECEIVER

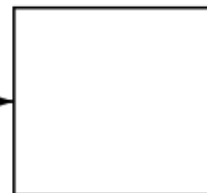
DESTINATION



SIGNAL

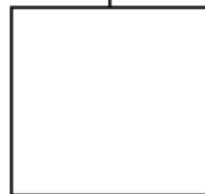


RECEIVED
SIGNAL



MESSAGE

MESSAGE



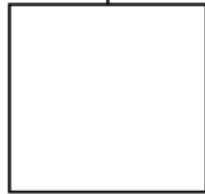
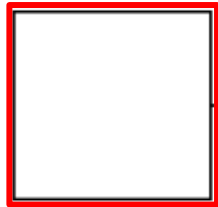
NOISE
SOURCE

INFORMATION
SOURCE

TRANSMITTER

RECEIVER

DESTINATION



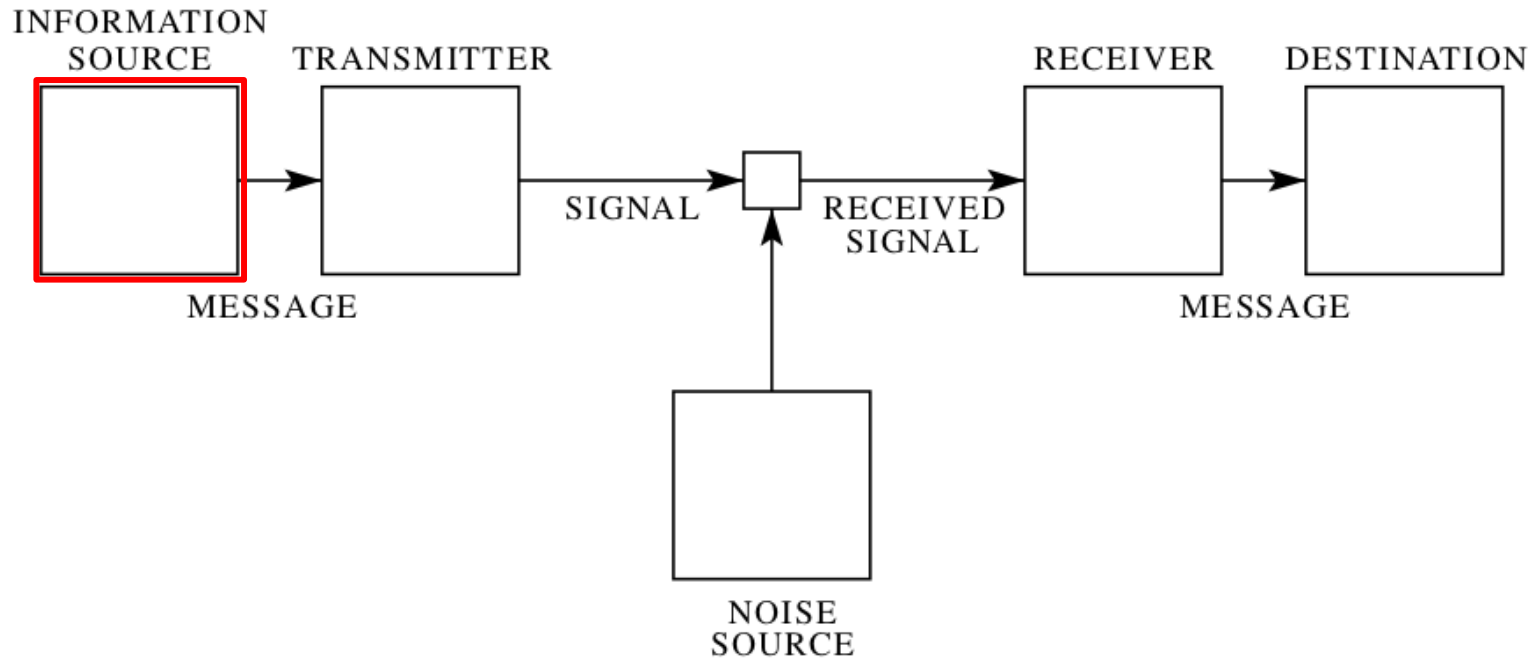
MESSAGE

SIGNAL

RECEIVED
SIGNAL

MESSAGE

NOISE
SOURCE

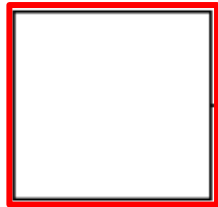


INFORMATION
SOURCE

TRANSMITTER

RECEIVER

DESTINATION



MESSAGE

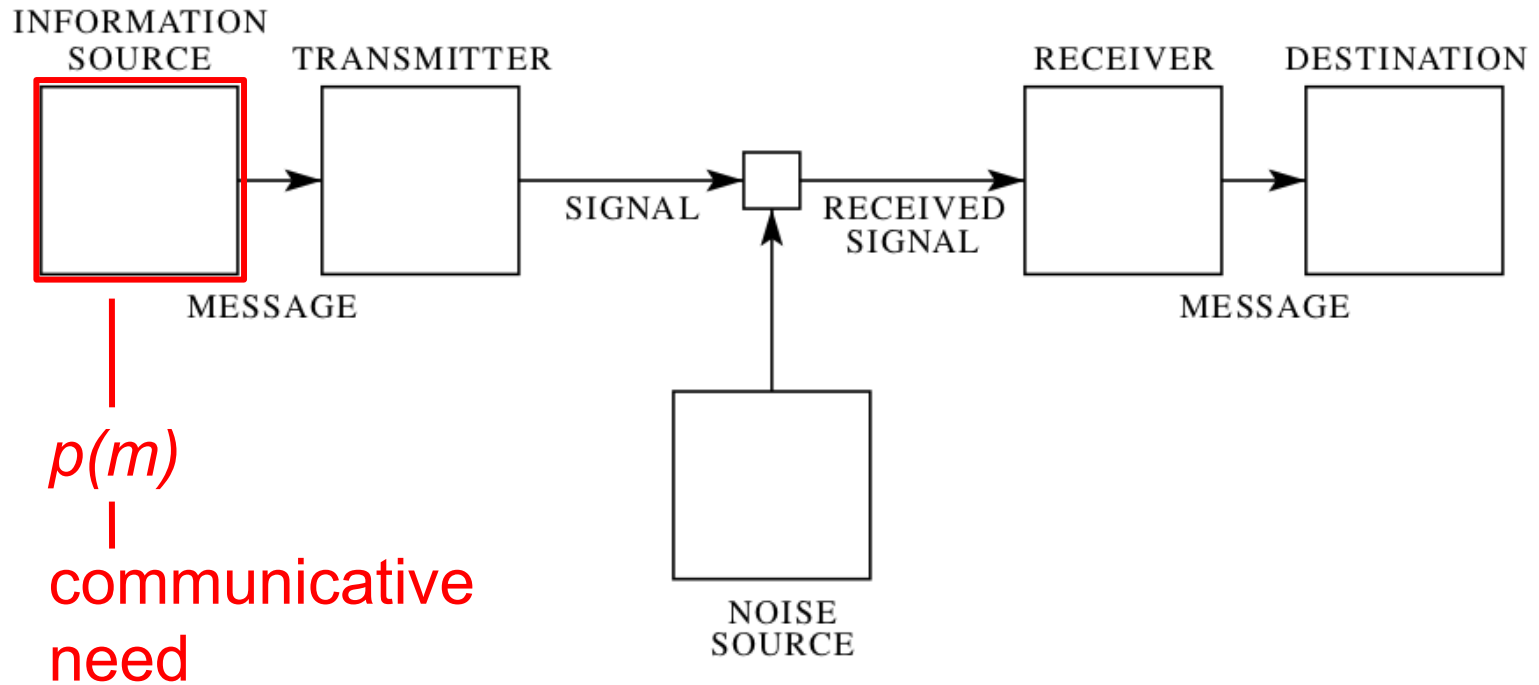
SIGNAL

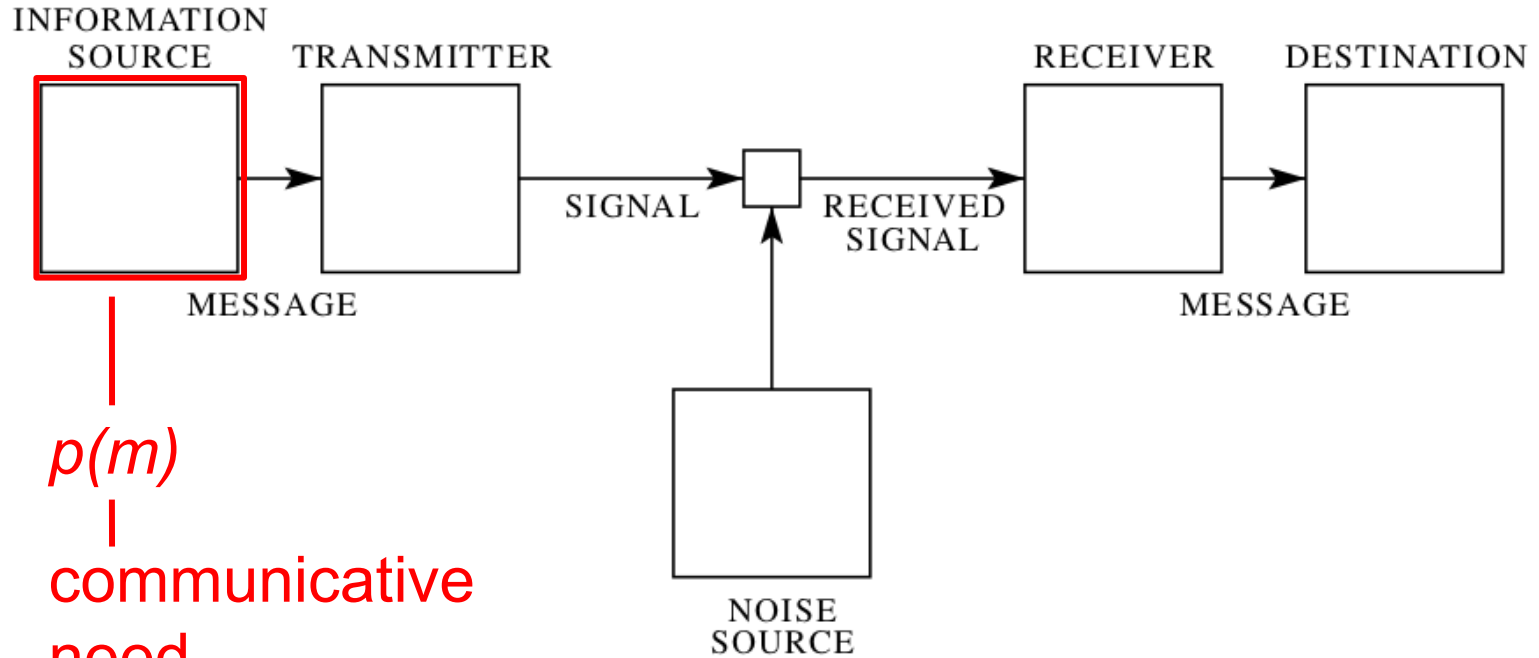
RECEIVED
SIGNAL

MESSAGE

NOISE
SOURCE

$p(m)$



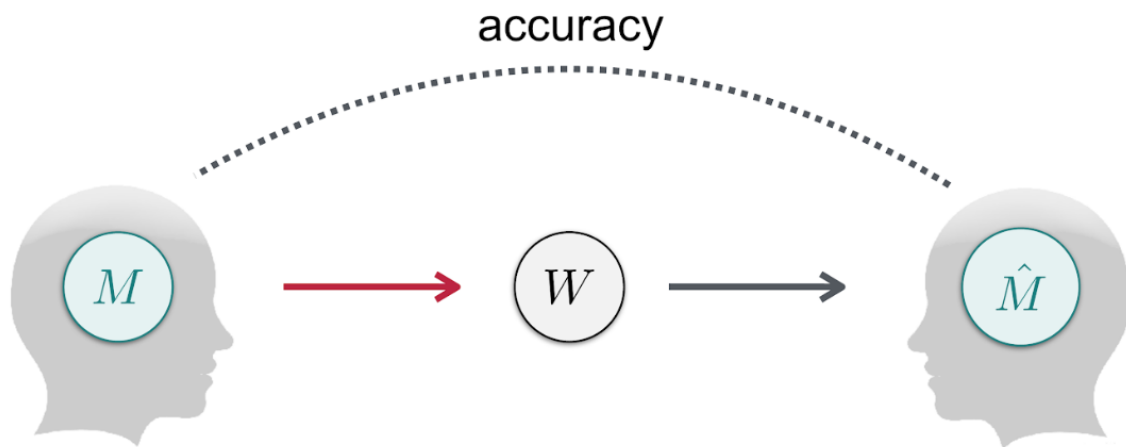


Potentially culturally determined:
“the chief interests of a people”

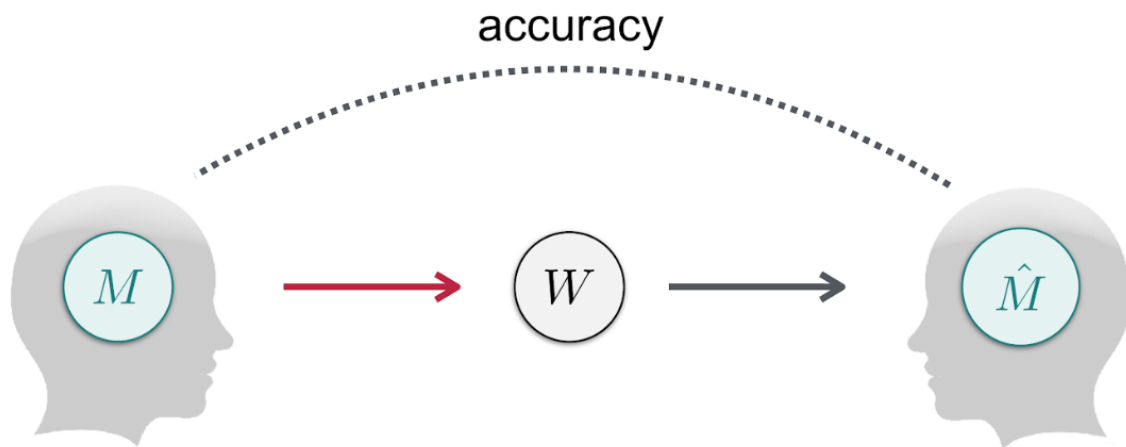
Is this in fact all “unredeemed piffle”?

Is this in fact all “unredeemed piffle”?

Or have we perhaps thrown the baby out
with the bathwater?

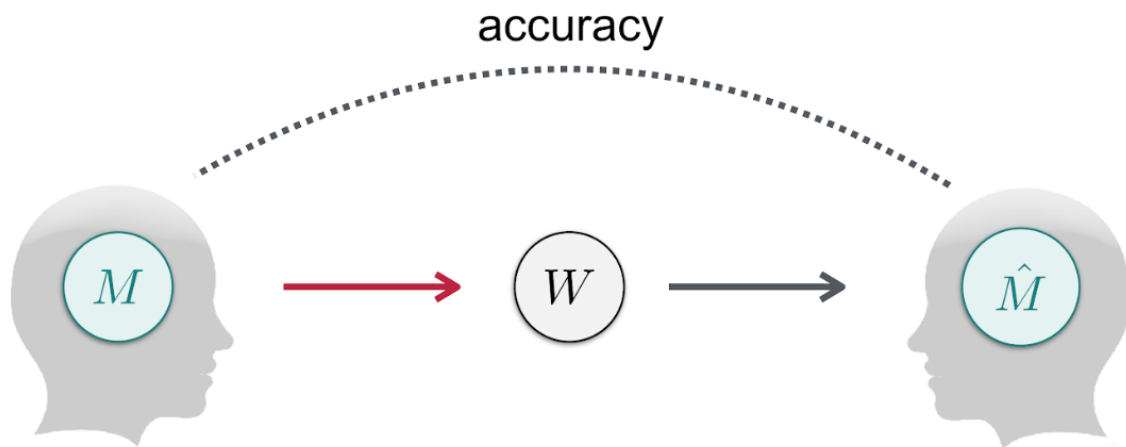


High accuracy means low distortion, $\mathbb{E}[D[M||\hat{M}]]$



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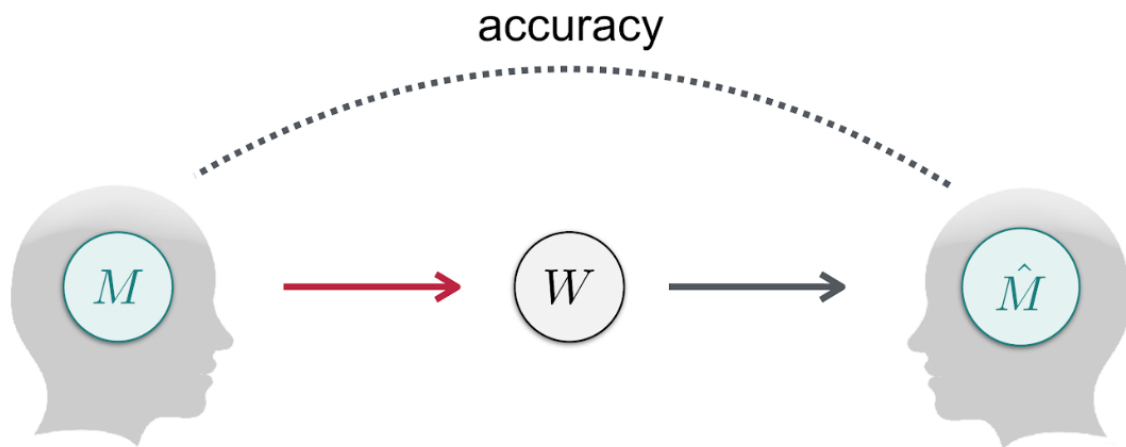
Penalizes
broad categories



High accuracy means low distortion, $\mathbb{E}[D[M||\hat{M}]]$

Weighted by frequency Penalizes broad categories

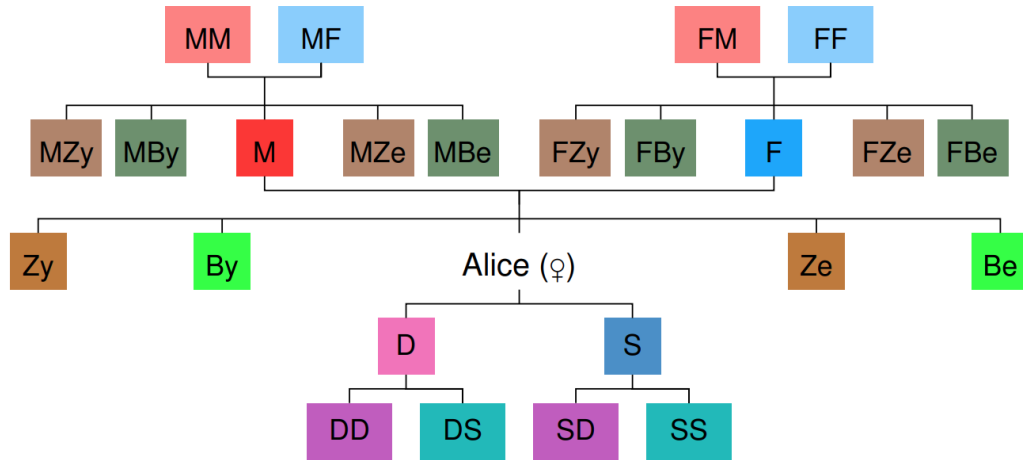
Fine-grained categories for frequently-referenced ideas



High accuracy means low distortion, $\mathbb{E}[D[M||\hat{M}]]$

Weighted by frequency Penalizes broad categories

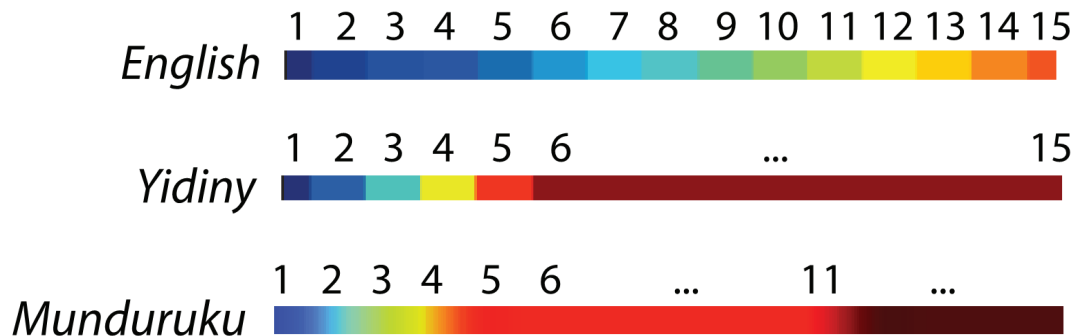
Fine-grained categories for frequently-referenced ideas



Kinship

Greenberg, 1966;
Kemp & Regier 2012

Fine-grained categories for frequently-referenced ideas



Numeral systems

Xu et al. 2020;
Pajot et al. 2025.

Fine-grained categories for frequently-referenced ideas

- These earlier analyses assumed a **universal** distribution of communicative need.

Fine-grained categories for frequently-referenced ideas

- These earlier analyses assumed a **universal** distribution of communicative need.
- What if communicative need were to **vary** across cultures and speech communities?

Fine-grained categories for frequently-referenced ideas

- These earlier analyses assumed a **universal** distribution of communicative need.
- What if communicative need were to **vary** across cultures and speech communities?
- We would expect to see the kind of pattern that Boas reported.

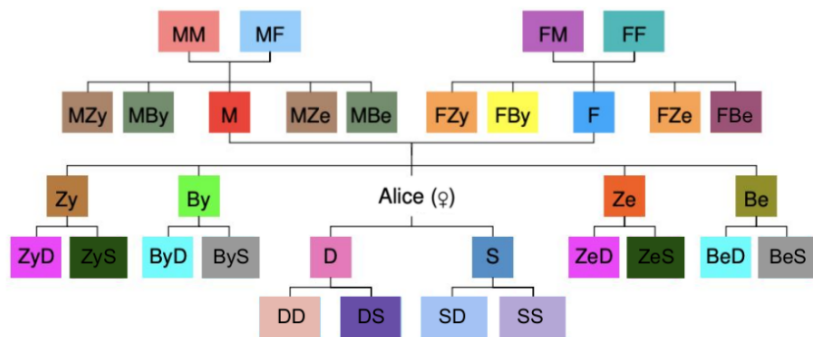
Kinship revisited



Anand & Regier, 2023, *Cognitive Science conference*

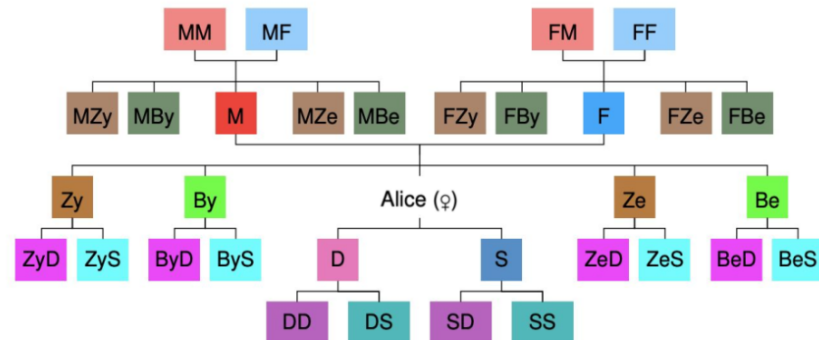
Kinship revisited

Hindi



- | | | |
|-------------------------------|--------------------------------------|----------------------------------|
| नानी (Maternal Grandmother) | ताऊ/बड़े पापा (Older Paternal Uncle) | बेटा/पुत्री (Daughter) |
| नाना (Maternal Grandfather) | (छोटी) बहन (Younger Sister) | बेटा/पुत्र (Son) |
| दादी (Paternal Grandmother) | (छोटा) भाई (Younger Brother) | नतिनी/नतिन (Daughter's Daughter) |
| दादा (Paternal Grandfather) | देदी/बड़ी बहन (Older Sister) | नाती (Daughter's Son) |
| मासी/मासी (Maternal Aunt) | भैया/बड़ा भाई (Older Brother) | पोती (Son's Daughter) |
| मामा (Maternal Uncle) | भांजी (Sister's Daughter) | पोता (Son's Son) |
| मम्मी/माँ (Mother) | भांजा (Sister's Son) | |
| पापा/पिता (Father) | भतीजी (Brother's Daughter) | |
| बुआ (Paternal Aunt) | भतीजा (Brother's Son) | |
| चाचा (Younger Paternal Uncle) | | |

English



- | | |
|-------------|---------------|
| Grandmother | Nephew |
| Grandfather | Daughter |
| Aunt | Son |
| Uncle | Granddaughter |
| Mother | Grandson |
| Father | |
| Sister | |
| Brother | |
| Niece | |

Anand & Regier, 2023, *Cognitive Science conference*

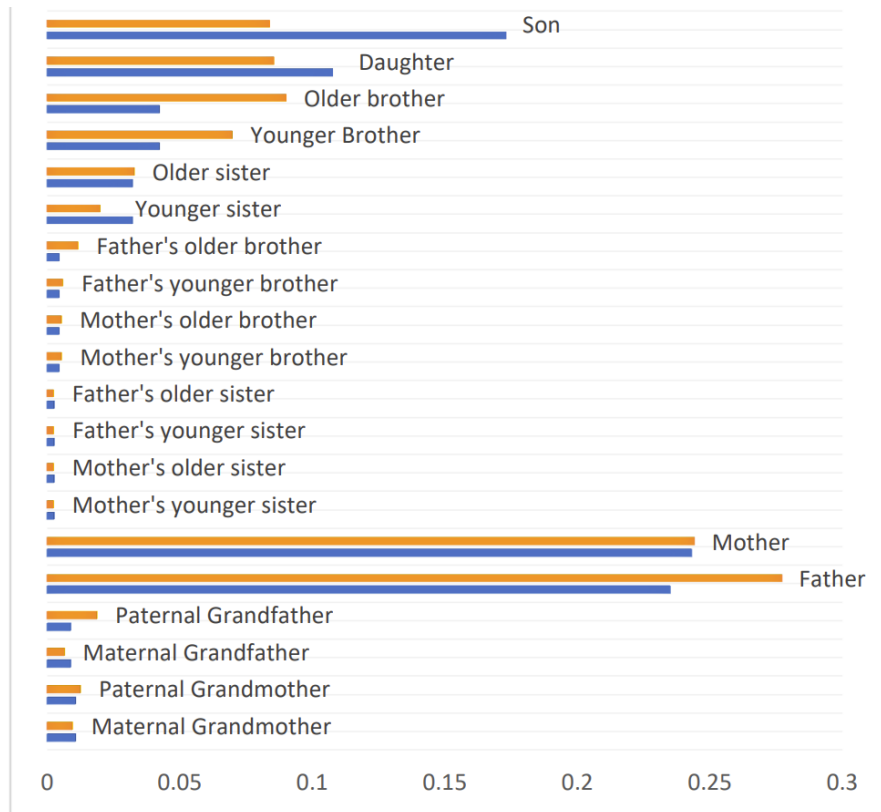
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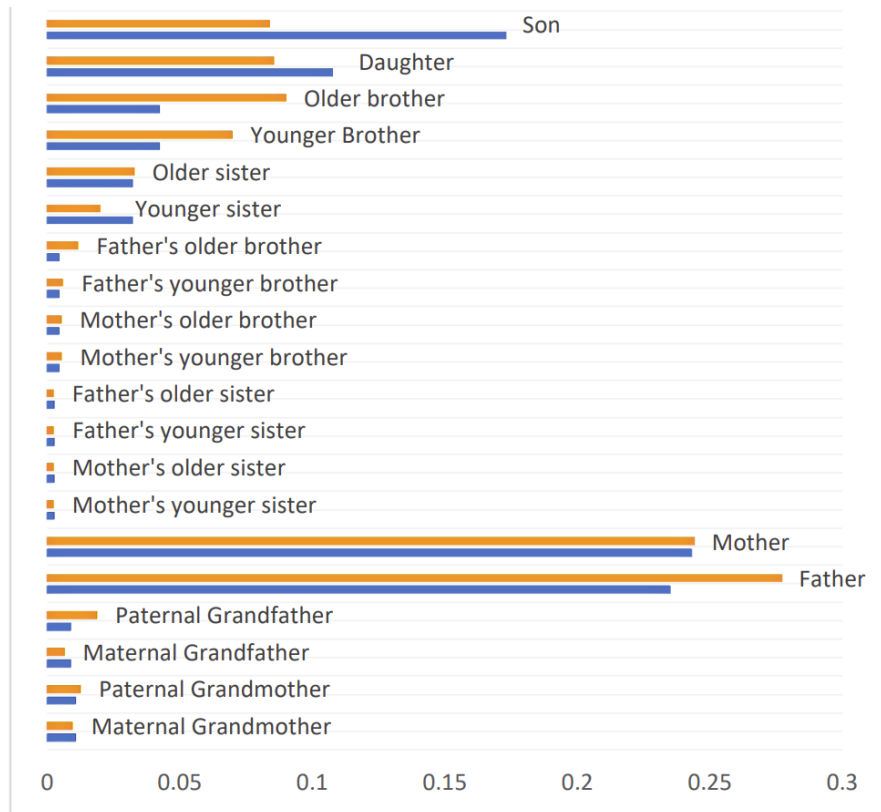
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- Greater **communicative need** to talk about kin in Hindi – consistent with finer-grained **naming**.

■ Hindi ■ English



Need probabilities $p(m)$ for kin types in Hindi and English, derived from Hindi and English web-scraped corpora.

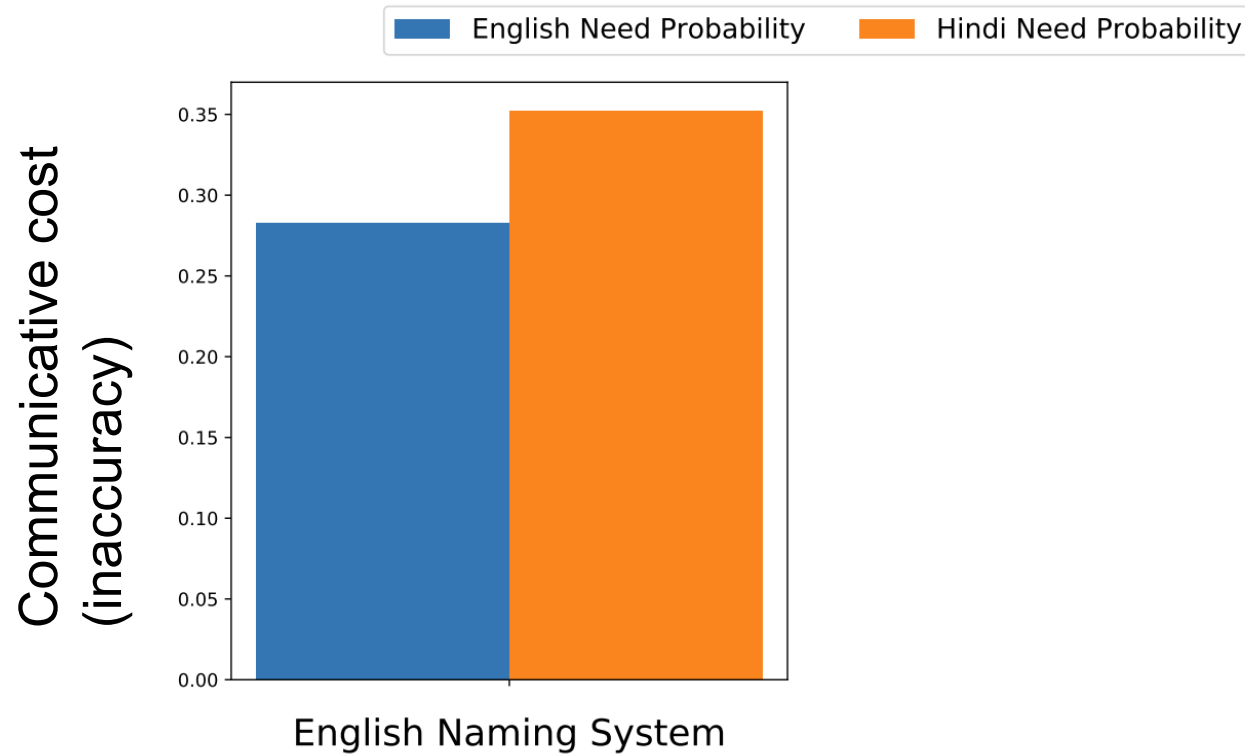
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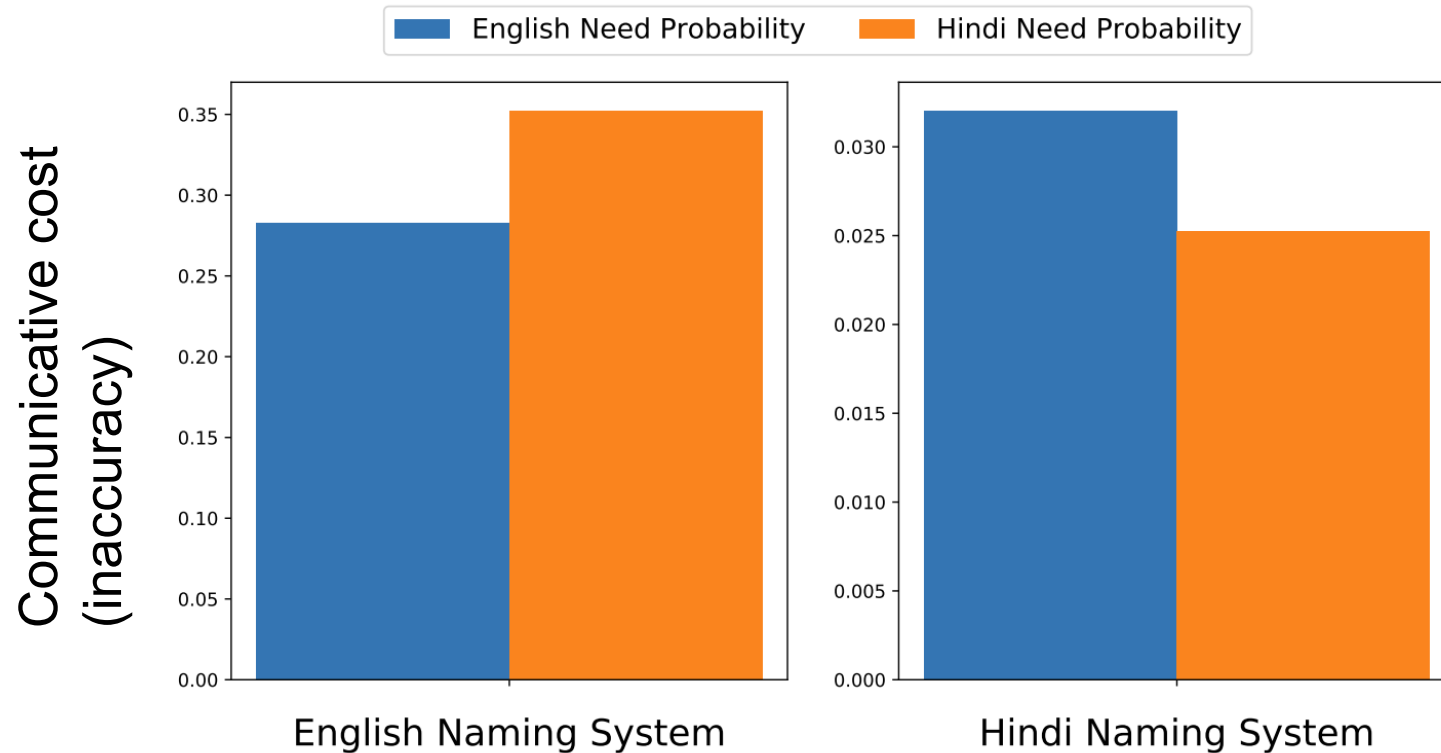
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Communication is **more informative** under **native-language** need probability.



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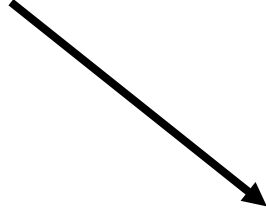




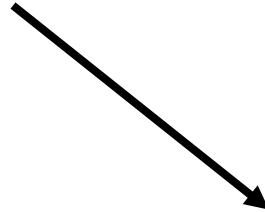


Single form for snow and ice
Nahuatl (Uto-Aztecan, Mexico)

Environment



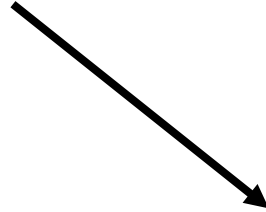
Communicative need



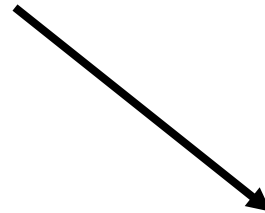
Category system



Environment

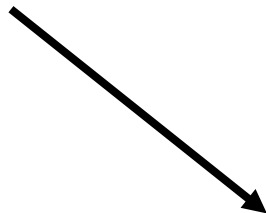


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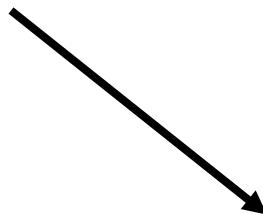


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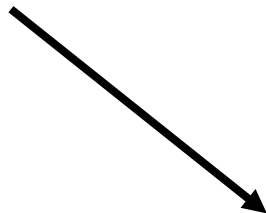


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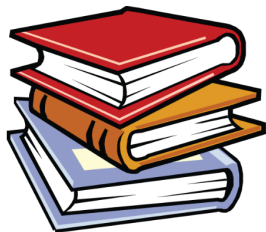
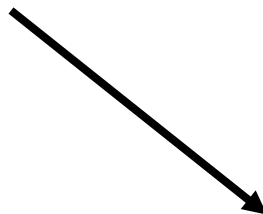


Category system

Environment



Communicative need



Category system

different terms for ice, snow



Regier, Carstensen, & Kemp, 2016, *PLOS ONE*

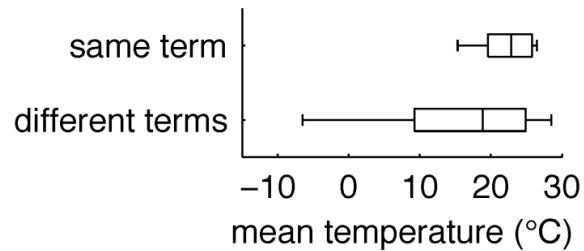
same term for ice, snow



different terms for ice, snow



Regier, Carstensen, & Kemp, 2016, *PLOS ONE*



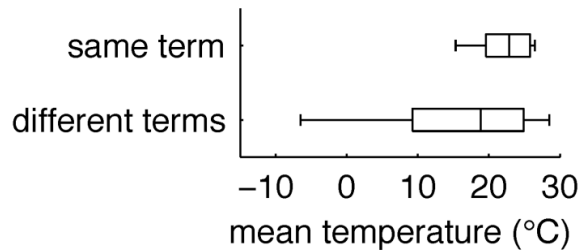
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Similar results from another cross-language dataset.

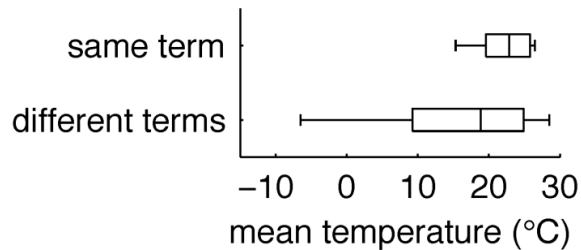
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Twitter analyses confirm that need probability for ice and snow drops off with mean temperature.

Regier, Carstensen, & Kemp, 2016, *PLOS ONE*



Single form for snow and ice
Arabic, Hausa, Hawaiian, Telugu, ...

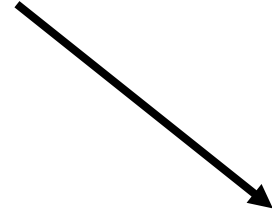
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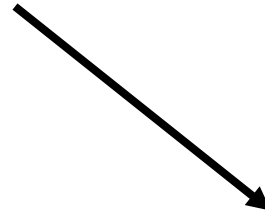
“languages that use the same linguistic form for snow and ice tend to be spoken in **warmer** climates, and ... this association appears to be mediated by **lower communicative need** to talk about snow and ice”

Regier, Carstensen, & Kemp, 2016, *PLOS ONE*

Environment



Communicative need



Category system

Cautionary note

- It is likely that need probability is **only part of the picture**.
- E.g. **need to distinguish**, or **discursive distinctiveness** also plays an important role in patterns of colexification (Beekhuizen, 2024).

Linguists Find Proof of Sweeping Language Pattern Once Deemed a ‘Hoax’

Inuit languages really do have many words for snow, linguists found—and other languages have conceptual specialties, too, potentially revealing what a culture values

Scientific American, May 9 2025



A computational analysis of lexical elaboration across languages

Temuulen Khishigsuren^a , Terry Regier^b , Ekaterina Vylomova^c , and Charles Kemp^{a,1} 

April 10, 2025





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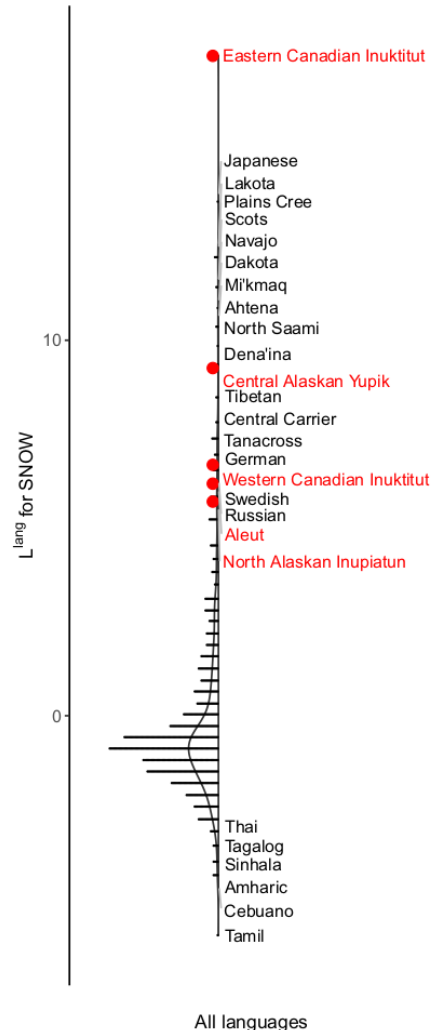


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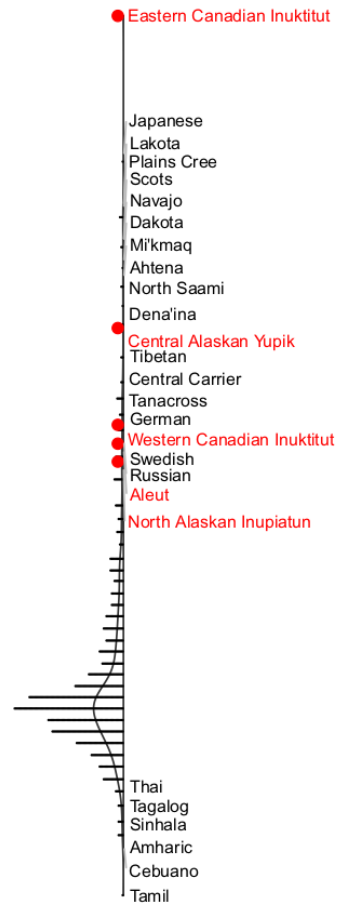
April 10, 2025

- Cross-language analysis of hundreds of **bilingual dictionaries** (X-English)
- **Inuit** dictionaries are among those that mention **snow** proportionally most often, suggesting many words related to snow.
- Other languages have **other conceptual emphases** (e.g. dance in N. American lgs)

$L_{\text{lang for SNOW}}$

10

0



Background

Background

- Everything I've said so far in this talk.

Definition

- **Lexical elaboration:** “The lexical elaboration of a concept within a language can be informally characterized as the **number of lexemes (or dictionary entries)** of that language that are **related** to the concept.”

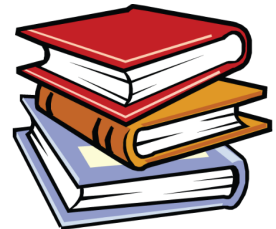
Definition

- **Lexical elaboration:** “The lexical elaboration of a concept within a language can be informally characterized as the **number of lexemes (or dictionary entries)** of that language that are **related** to the concept.”
- NB: not the same as **subtypes** of a concept: **snow**-related includes **falling snow** (a type of snow) but also **snowshoes** (related to snow but not a type of snow).

Novel contributions

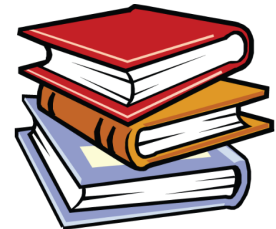
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- Use BILA to **test independent claims** about lexical elaboration.

E.g. “Mongolian has many terms related to horses”

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- Use BILA to **test independent claims** about lexical elaboration.
- Explore how lexical elaboration **correlates** with **ecological** and **cultural** variables.

Examples of lexical elaboration drawn from dictionaries in BILA

Concept	Language	Term	Definition
SNOW	Inuktitut (5) Eskaleut / Inuit: Canada	<i>aput</i>	Snow (spread out)
		<i>auverk</i>	Snow block (for building)
		<i>panar</i>	Snow knife
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DANCE	K'iche' (11) Mayan: Guatemala / Mexico	<i>xahoh</i>	To dance
		<i>xahovizah</i>	Cause to dance, invite to dance, play with a dancing toy, such as a top
		<i>tu</i>	Dance on stilts
		<i>Abrojo</i>	A clown in the dance of the Conquest
		<i>er</i>	Sound made by the Mute Boy in the dance of that name
		<i>Convite</i>	A widely popular ritual dance in which one invited "couple" from each of several towns will be represented in municipal costume; commonly comic, with speeches in honor of the saint of the day; all the dancers are male, though women are personified

L^{lang} for SNOW:



L^{lang} for SNOW

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L^{lang} for SNOW:

A measure of how
lexically elaborated the
concept SNOW is in a
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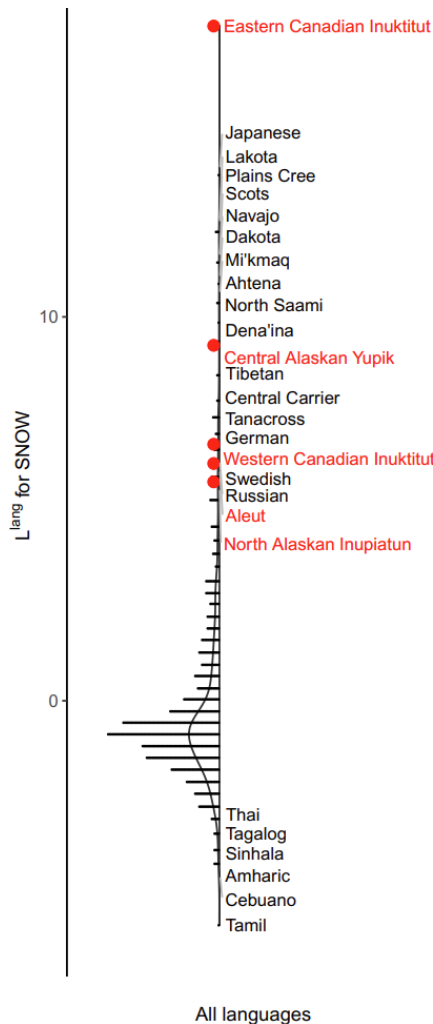
L^{lang} for SNOW

The 5 Eskimo-Aleut languages in BILA are shown in red.

L^{lang} for SNOW:



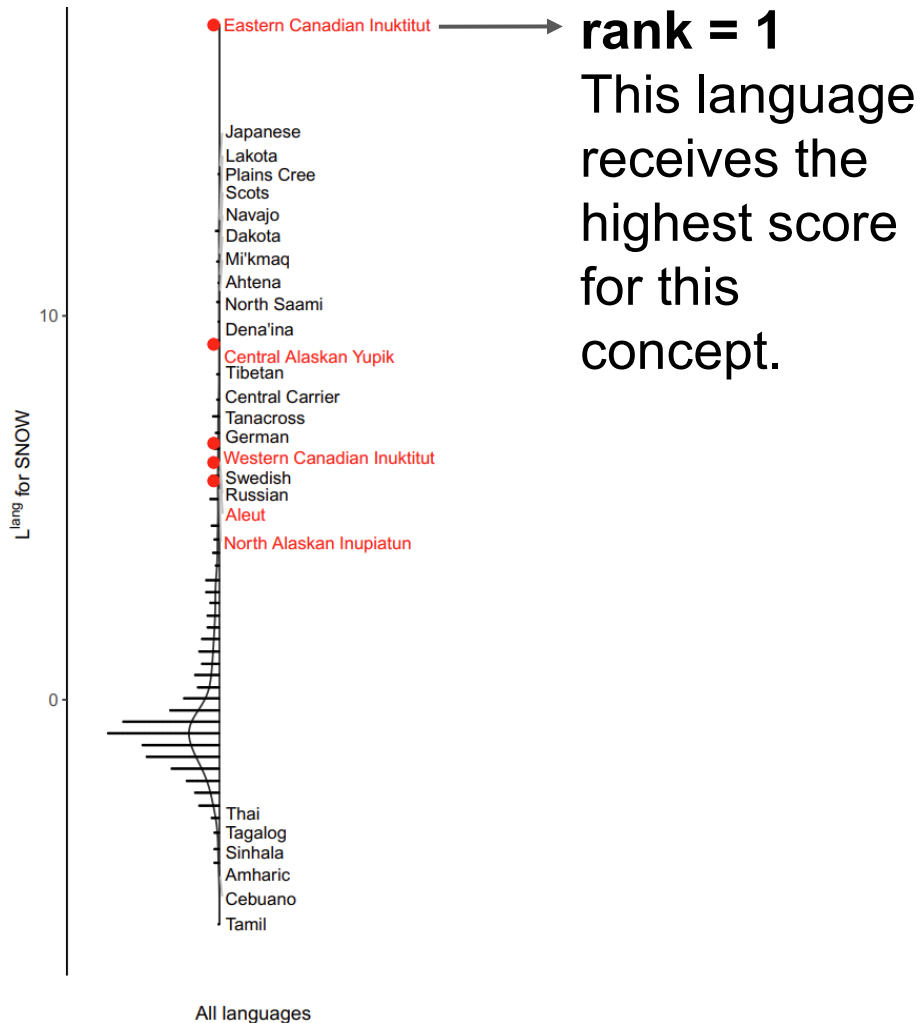
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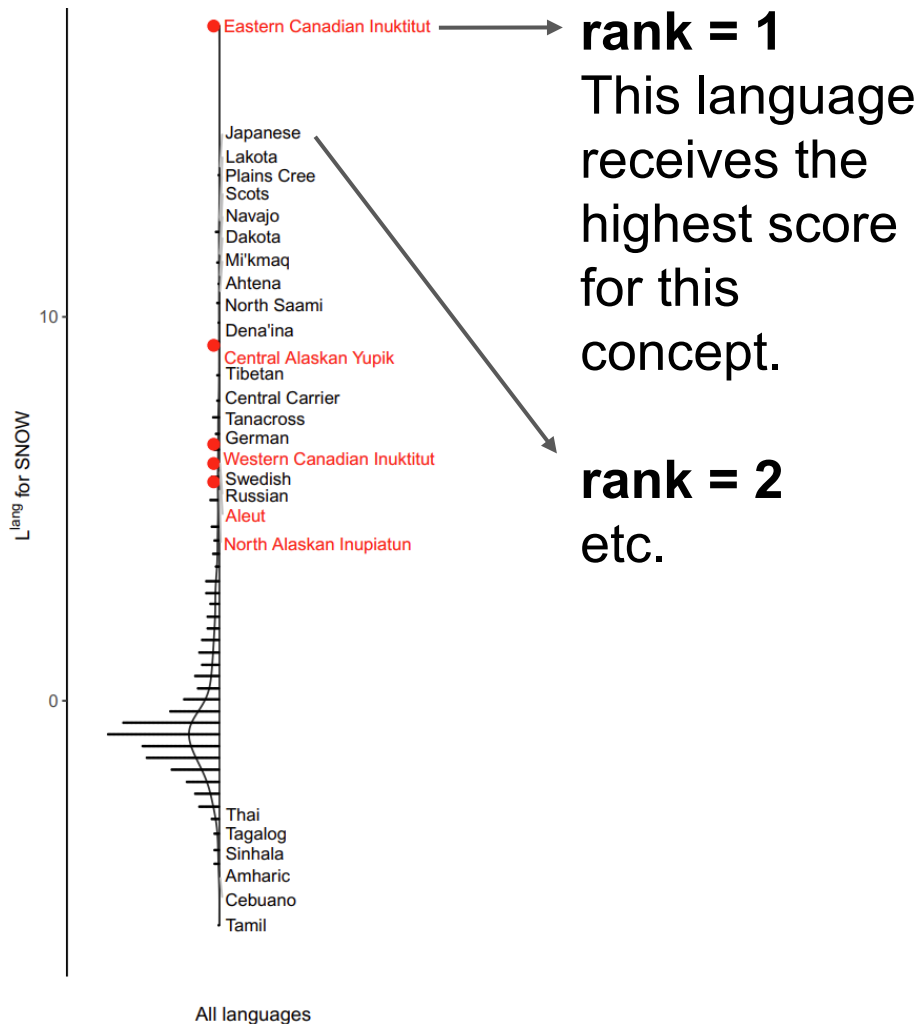
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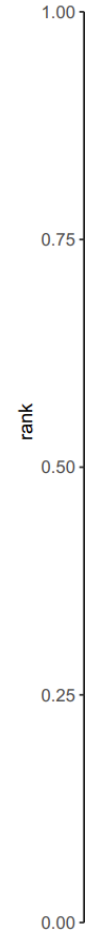
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**Assessing prior claims about lexical
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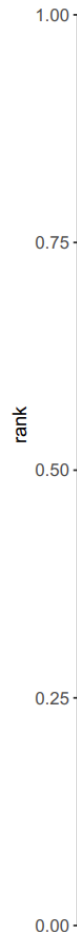


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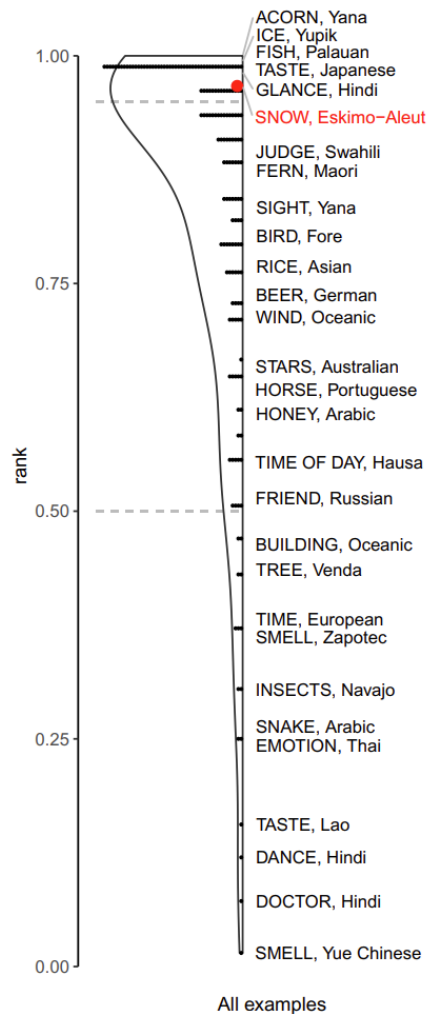
For each (language, concept) pair for which **a claim has been made**, the rank of that language with respect to that concept.



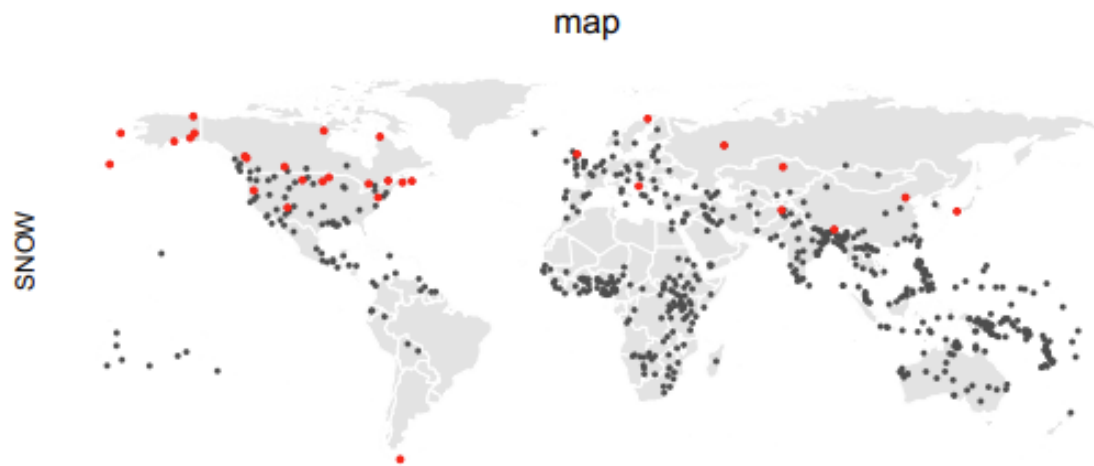
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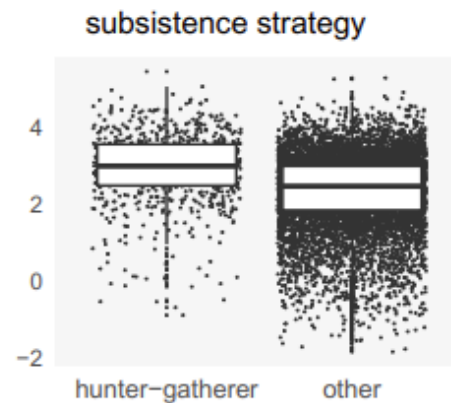
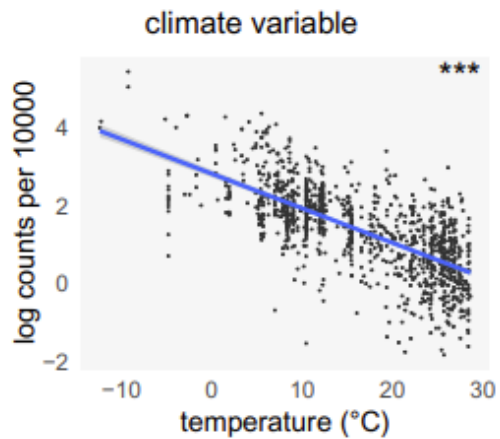
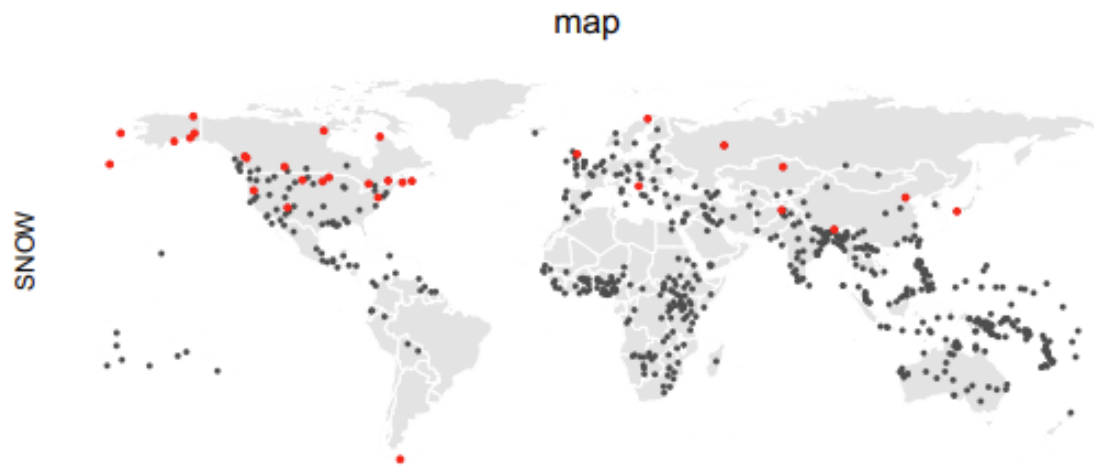
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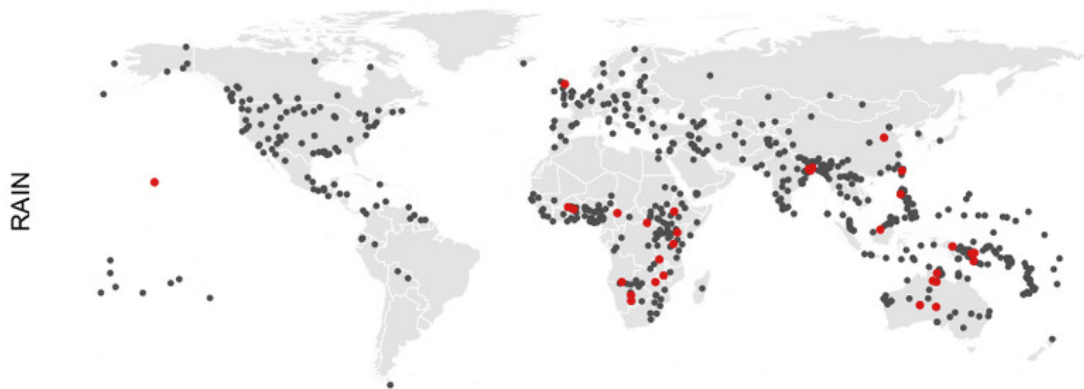
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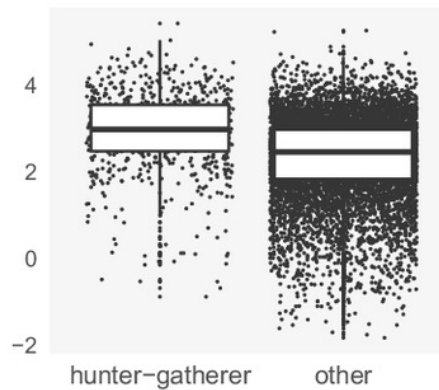
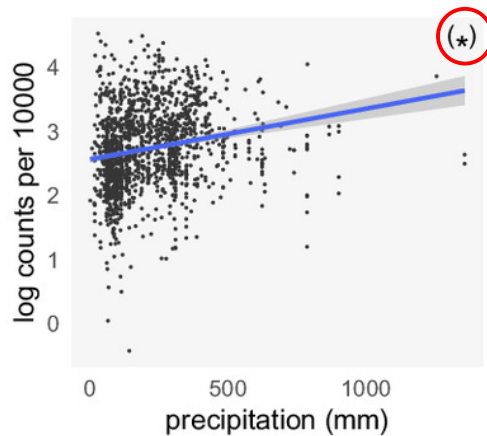
RAIN



RAIN



Does not survive robustness checks



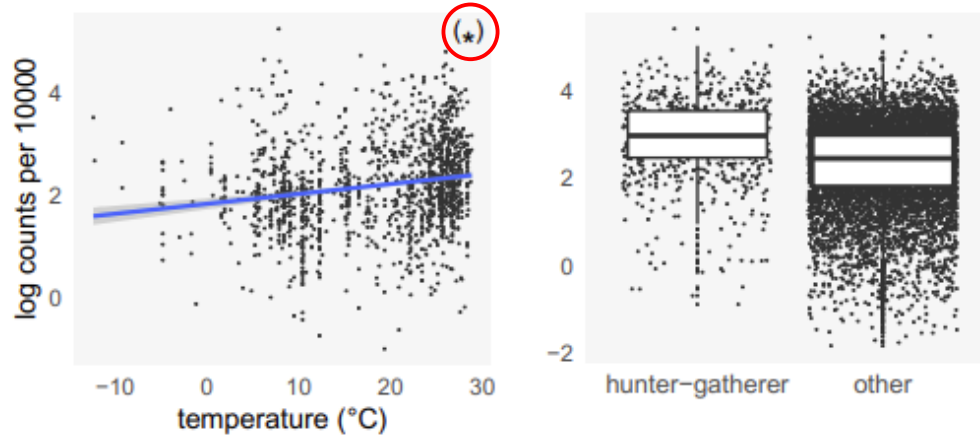
DANCE



DANCE



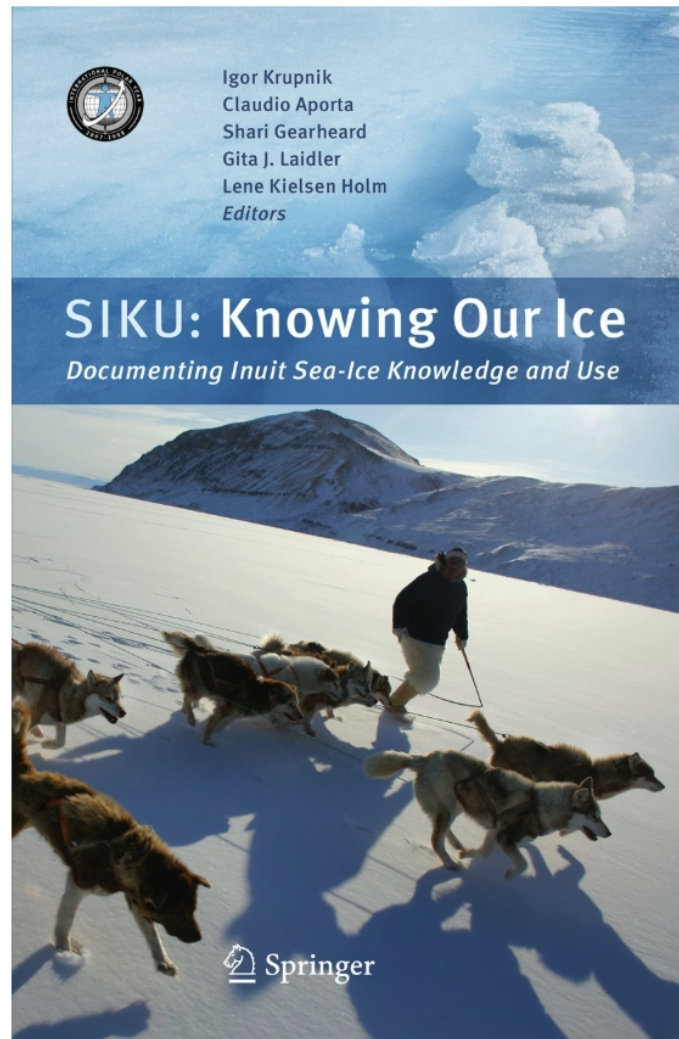
Does not survive robustness checks



One last example, from:

I. Krupnik & L. Müller-Wille
(2010). Franz Boas and
Inuktitut terminology for ice and
snow: From the emergence of
the field to the “Great Eskimo
Vocabulary Hoax”.

In this volume 



Russian *protalina*:
“open ground where
the snow has
melted”

Krupnik & Müller-Wille 2010,
p. 394



Let's not throw out the baby
with the bathwater!

Conclusions

- Both Boas and Shannon have legacies that touch the question of semantic categories, with some **tension** in each case.

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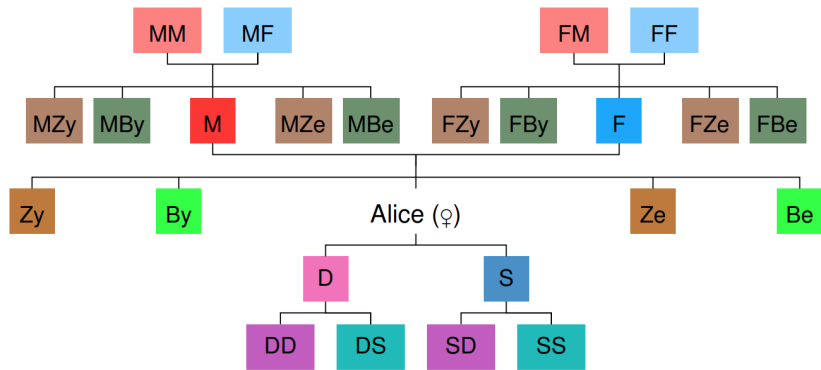
Conclusions

- Both Boas and Shannon have legacies that touch the question of semantic categories, with some **tension** in each case.
- Considering the two **together** can help to **resolve** some of that tension.
- Situates a **classic proposal** about language and culture in information-theoretic terms.

Thanks to my **collaborators.**

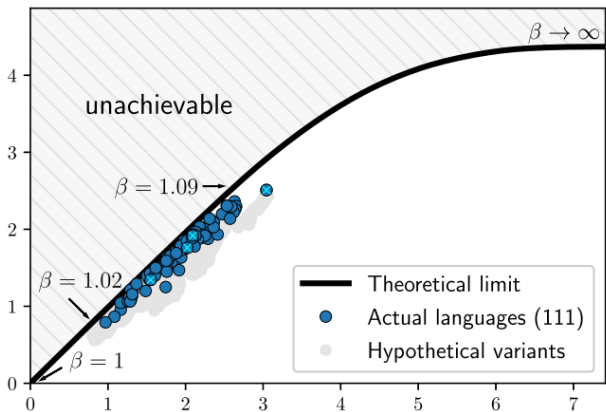
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Thanks to the many **lexicographers** and **fieldworkers** on whose work we rely, and especially the **speakers** who ultimately supplied the data.

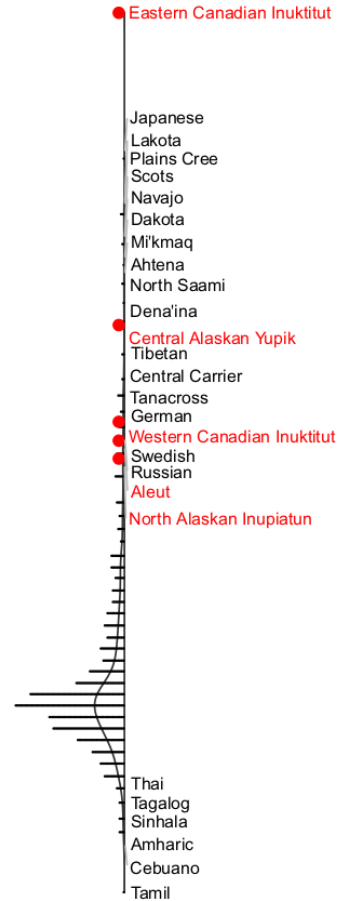


$p(m)$

Questions?



$L_{\text{lang for SNOW}}$



All languages

Course evaluations

Please take a few minutes to fill out your
course evaluation!